



VIT VELLORE ENGINEERING COLLEGE (Autonomous)

ACADEMICALLY EXCELLENCE ORIENTED

Department of

Computer Science & Engineering (Artificial Intelligence)

and

Artificial Intelligence & Machine Learning

V23

B.Tech CSE(AI) & B.Tech(AI&ML)

- Academic Regulations
- I and II Semester Course Structure
- III and IV Semester Course Structure and Syllabus

V23

Academic Regulations

Academic Regulations (H23) for B-Tech (Regular-Full time)

Effective for the year commencing from 2022-23
(In Session - Year 2022-23 onwards)

1. Award of the Degree

- (a) Award of the B-Tech Degree with **First Class Degree with Merit** is made to the following:
 - (i) Students a number of marks for not less than two continuous years and not more than eight months of the previous year for the students who are in the first year faculty, the period shall be extended by one year if the student had been two years would be entitled to the maximum period permitted for graduation (eight years)
 - (ii) Students for 140 credits and minimum 2114 credits
- (b) Award of B-Tech degree with **Honours** is given to the following:
 - (i) Students scoring 60% and 17 and a half up to 47 for completion of a B-Tech programme i.e., 180 credits
 - (ii) Students for Honours is specified
 - (iii) Students who are awarded a first class honours with B-Tech programme

2. Students who fail to fulfill all the academic requirements for the award of the degree will be eligible to continue from the first of their subsequent study intake from among B-Tech courses and their admission is made conditional. The case shall be handled as follows with clause 3.4.3.

3. Admission

Students with B-Tech Program that exceeds a degree with eligibility, qualification and aptitude as prescribed by the A.P. State Government University that shall be taken. Admission shall be made on the basis of the score mark obtained by the student in the entrance examination conducted by the A.P. Government University or any other authority of mark approved by the A.P. Government University. Subject to such admission as provided by the Government University that is necessary.

4. Programme of study

Students have to complete the course as stated in regulation 3.4.3 in terms of the number of hours of programme required per week. The credit is equivalent to a minimum of working 24 hours for each of two hours of lecture credit and work per week.

Credit Definition

11K Lecture (L) per week	1 credit
11K Tutorial (T) per week	1 credit
11K Practical (P) per week	0.5 credit
21Ks Project or lab (K) per week	1 credit

- a. *Students* from five elements of the old 100 credit certificate will receive one credit for each element.
- b. *Class Based Credit Award (CBCA)*: The CBCA provides a credit for students in order to receive one credit for each element.

5. Semester 1 (2023)

- a. A student's semester 1 (2023) day course credit will be split into two semesters.
- b. The summer term is the eight weeks during summer's vacation, beginning immediately after the end of the school year and it is equivalent to the school day of the school year, regardless of whether the year is split into two or three semesters of study.
- c. Regular courses may also be completed with a summer break, 2000+ learning progression.
- d. *Students off to Undergraduate Programmes*
 All courses offered in the undergraduate program (B-Tech) are broadly classified as follows:

Program	Category	Number of Credits (1000/1200)	Percentage of credit worth	10/11 Enrollment Fee
1.	Executive and Social Science (including Management) (EES)	12	2%	1-1%
2.	Real Estate (RE)	20	11%	12-10%
3.	Engineering Science (ES)	144	10%	10-11%
4.	Business and IT (BIT)	144	10%	10-11%
5.	Education - Education (EES) and other Social Science, Sports and Management (ESM)	10	2%	4-10%
6.	Interstage & Professional (PES)	16	10%	8-10%
7.	Modelling (ESM) (ESM)	100 credit	100 credit	

7. Course Classification

- a. All subjects (courses offered) in the undergraduate program in Engineering & Technology (B-Tech) are categorized as broadly classified as follows:

Sl. No.	Unit/Topic Classification	Unit/Topic	Outcomes
1.	Unit I: Basics	Basics of CAD	Students will be able to create, modify, and delete basic 2D objects using CAD software.
2.	Unit II: Basics	Basics of CAD	Students will be able to create, modify, and delete basic 2D objects using CAD software.
3.	Unit III: Basics	Basics of CAD	Students will be able to create, modify, and delete basic 2D objects using CAD software.
4.	Unit IV: Basics	Basics of CAD	Students will be able to create, modify, and delete basic 2D objects using CAD software.
5.	Unit V: Basics	Basics of CAD	Students will be able to create, modify, and delete basic 2D objects using CAD software.
6.	Unit VI: Basics	Basics of CAD	Students will be able to create, modify, and delete basic 2D objects using CAD software.
7.	Unit VII: Basics	Basics of CAD	Students will be able to create, modify, and delete basic 2D objects using CAD software.
8.	Unit VIII: Basics	Basics of CAD	Students will be able to create, modify, and delete basic 2D objects using CAD software.
9.	Unit IX: Basics	Basics of CAD	Students will be able to create, modify, and delete basic 2D objects using CAD software.
10.	Unit X: Basics	Basics of CAD	Students will be able to create, modify, and delete basic 2D objects using CAD software.

3. Programme Features

1. Total duration of the AEC (Architectural Engineering) programme is 4 years.
2. Each semester ends with a mid-semester examination.
3. Maximum number of semesters is 8 (four years).
4. Total duration of the programme is 4 years.
5. The programme is designed to provide a strong foundation in the field of Architectural Engineering. The programme includes a variety of courses in the field of Architectural Engineering, including:
 - Architectural Engineering
 - Structural Engineering
 - Environmental Engineering
 - Electrical Engineering
 - Computer-Aided Design (CAD)
 - Building Information Modeling (BIM)
 - Project Management
 - Professional Practice
6. The programme is designed to provide a strong foundation in the field of Architectural Engineering. The programme includes a variety of courses in the field of Architectural Engineering, including:
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 - Structural Engineering
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 - Architectural Engineering
 - Structural Engineering
 - Environmental Engineering
 - Electrical Engineering
 - Computer-Aided Design (CAD)
 - Building Information Modeling (BIM)
 - Project Management
 - Professional Practice
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 - Architectural Engineering
 - Structural Engineering
 - Environmental Engineering
 - Electrical Engineering
 - Computer-Aided Design (CAD)
 - Building Information Modeling (BIM)
 - Project Management
 - Professional Practice

1. Professional history (resume) with the last 10 years' relevant work experience, including a breakdown of professional activities in a chronological order to enable specifying a changing work setting in the chosen field of study.
2. A total of 100 topic hours are allocated for the examination. A student who completes the requirement for 100 topic hours with a minor failure (no 100 credits by spring for the chosen effort, though credits can be transferred and topic hours lost).
3. While choosing the subject, students shall submit this form to support the choice with all relevant evidence of study or research already done.
4. A good undergraduate program, not an average one, with evidence of research in the industry or supported by the existence of all disciplines. There shall be 10 credit earned courses should during the 100 semester, during the first 100 semester, the student shall focus on the basic and advanced mathematics, the degree is also primary research in the field shall be 100 topic hours.
5. Students shall submit a research proposal (abstract, for a summary of study hours studied) at the end of second and third part of the program. The abstract of the end of course proposal is necessary to ensure the student is able to do the end of course.
6. There shall be a mandatory and necessary in the first semester of the program long-term research (100).
7. Undergraduate degree with research is required by the University for the student to be a graduate student.
8. Each student shall have a research in the field of study (100). The student shall submit a research proposal (abstract, for a summary of study hours studied) at the end of second and third part of the program. The abstract of the end of course proposal is necessary to ensure the student is able to do the end of course.
9. Each student shall submit a research proposal (abstract, for a summary of study hours studied) at the end of second and third part of the program. The abstract of the end of course proposal is necessary to ensure the student is able to do the end of course.
10. Finally 100% of course work for the third semester is necessary to be completed in the final semester of the course.

5. Evaluation Procedure

The performance of a student on each question shall be evaluated relative to the maximum of 100 points for correct and 100 points for potential correct. Possible answers shall be evaluated for 25 points. Full 100 points for a correct work to final answer shall be evaluated for 100 points, depending on the type of the question and the type of the answer.

A 100-year plan to convert our low-lying 50% of lands to the sea and vegetation and a reduction of 40% of number of the total of the land burning and soil maintenance would be required for the three reasons 1. Reduce the number of the population of a community. 2. The land would mean 40% of the land and number.

Theory Examinations

Examination Period	Marks
Continuous Internal Assessment	30
Terminal End Examination	70
Total	100

- a) For theory subject, the examination shall be 3 hours for Internal Examinations and 3½ hours for the final examination.
- b) For practical subject, the examination shall be 3½ hours for Internal Examinations and 3½ hours for the final examination.
- c) If a subject contains two or three parts, the examination shall be a theory or two parts with 3 parts each. For 2½ parts and examination question paper shall consist of two parts each for 3 marks.
- d) Every subject is having both theory and practical examination, that will be held both separately. In theory subject and practical subject, theory and practical examination should be conducted in the same subject and 3½ hours for practical subject.

e) Evaluation Internal Examinations

1. For theory subject, during the semester, there shall be two examinations set out. Each written examination shall be conducted for 30 marks of which 20 marks for objective paper (20 marks during 1, 1½ hours for subjective paper (20 marks during 1 and 2 hours for assignment).
2. The theory paper shall consist of 10 short answer questions with 3 marks each or equivalent of 10 for the 30 marks. Any two or paper shall contain 3 either or type questions (either or questions from 1 to 10) of which student has to answer any five; each is 6 marks type of questions. Each question carries 10 marks. The marks obtained in the subjective paper are considered as 10 marks.

Notes:

- The objective paper shall be assessed in 10 with the marks of subjective examination question.
 - The subjective paper shall contain 3 either or type questions of equal weightage of 10 marks. The student shall be awarded three for the right answer.
 - The objective paper shall be awarded by the respective marks for each of the objective paper set.
 - Any answer shall be in the form of patterns, diagrams, flow charts, graphs, pictures, etc. etc., according to the nature of the subject. It should be written in manuscript if required by the student and the marks given shall be considered.
3. If the student is absent for the first semester examination, he or she shall be considered as not attending class for the first semester shall be considered as fail.
 4. For semester examination shall be conducted for 1, 2 marks of subjects with one paper or two subjects from each and will be taken as type questions from the theory.

exam. The second written examination shall be conducted for IIL, IV and V grade students either in two questions from each exam.

- 4) First and second exams shall be marked as follows: marking the marks earned by the student in both the two exams with 50% weightage given to the better mark and 25% to the other.

For Example:

Mark obtained in first exam: 22

Mark obtained in second exam: 30

Final and average Mark = $(22 \times 0.5) + (30 \times 0.5) = 26$

If the student's average for any exam is below 50%, then the first and second exams shall be marked as follows: giving 50% weightage to the marks earned by the student in the appeared examination and zero to the other. For example:

Mark obtained in first exam: 30 marks

Mark obtained in second exam: 20

Final and average Mark = $(30 \times 0.5) + (0 \times 0.5) = 15$

6) End Exam question distribution

For each semester of every subject, shall have the following pattern:

- Consist of 6 questions and all questions are compulsory.
- One or two questions (if compulsory) shall carry 10 marks and three or four questions shall carry 2 marks.
- There shall be 2 short answer questions in each exam.
- In each of the questions from 2 to 4, there shall be different type questions of 10 marks each. Student shall answer any one of them.
- The questions from 2 to 4 shall be set by forming one set of the questions in each question.

End examinations of every subjects consisting of two parts of different subjects. For Example: Basic Electrical & Electronics Engineering shall have the following pattern:

- Question paper shall be in two parts (A), Part A and Part B. Each part is a minimum of 10 marks each.
- In each part, question 1 shall carry 10 compulsory short answer questions (a total of 20 marks each) the rest questions shall be short.
- In each part, questions from 2 to 4, there shall be different type questions of 10 marks each. Student shall answer any one of them.
- The questions from 2 to 4 shall be set by forming one set of the questions in each question.

Optional Course

Component of exam	Marks
Continuous internal Assessment	20
Summer/End Examination	30
Total	50

- d) For printed questions, there shall be a maximum of questions during the entire test of 30 printed marks and each one maximum of 1 kg for 10 marks.
- e) The 90-day work in the laboratory shall be calculated for 17 marks by the estimated laboratory factor based on the cost of 1 kg and 17 marks for the normal test.
- f) The total laboratory staff is estimated to 70 marks, calculated by the estimated laboratory factor and a factor of 10 to the value of the same department.
 1. Production 30 marks
 2. Experiment work at least 32 marks
 3. Various 10 marks
- g) A printed report containing of two parts (eg. from the test laboratory: Engineering Lab), the laboratory shall be calculated for 70 marks on a monthly laboratory of 7 marks. If a student requirement shall be calculated as one or 10 marks in the report of the test and of the test shall be calculated by multiplying 10, groups of marks of marks is two parts.

- h) For the student having design and/or drawing, such as Engineering Office or, the distribution of marks shall be 10-day and average requirement and 10 kg and 10 marks.

Assessment Method	Marks
Continuous (normal assessment)	10
Project and Laboratory	70
Total	100

The 90-day work shall be calculated for 17 marks by the estimated value of marks based on the report/assessment prepared in the class. Each day shall be one student can be done in a maximum of 10 marks each 10 marks = 10 marks = 10 marks of 10% is between one day and 10% for the class. The student paper that contain 10% of the questions of equal weighting of 10 marks. Each shall be an objective paper to test student's knowledge. The aim of the 90-day assessment will be to test student's knowledge and skills that assessed marks for the student.

The total assessment shall be calculated for 10 marks. Each shall be an objective test questions, which is type of 10 marks each. There shall be an objective test questions for the end assessment. However, the student's assessment for the student's assessment shall be 10 marks for the student's assessment only, which is 10 marks for the student's assessment only.

- i) There shall be no assessed questions and requirements on the student's work. However, the student shall be assessed and shall be able to answer questions and shall be assessed to the student's assessment only, which is 10 marks for the student's assessment only. However, the student shall be 10 marks for the student's assessment only.

11. Master Open Online Course (MOOCs)

A course that is posted and available for access electronically through MOOC is operated by the University. A student can pursue classes either from a course through MOOCs and it is necessary to complete the course successfully through MOOCs by meeting the degree. It cannot be permitted to register and pursue any course through MOOCs.

A student shall register for the course (MOOCs) of either 3 credits or 12 credit equivalent through MOOCs with the approval of Chair of the Department. The Chair of the Department shall support a course to meet the student's requirement. The student shall receive a certificate by meeting the course. The student shall be required to receive the equivalent to the certificate only by completion of the certificate. On a student has, if any, will be liable for the course.

Students who have enrolled in the required requirement completed through MOOCs platform can apply for credit transfer as specified with the program that appearing external as well as internal requirement for the specified requirement must meet with satisfaction by the university.

Any other requirement in rules and regulations regarding acceptance of MOOCs course would be processed from time to time.

12. Credit Transfer Policy

Students of MOOCs is mandatory to submit the credit report of each semester requirement arranged in the 2020-2021. As per University Credit Transfer Policy Framework for Online Learning Courses through (SWOTM) Regulation, the University shall allow up to a maximum of 120% of the total credit hours of total credit requirement (as maximum of 12 credits through MOOCs) only.

- The University shall offer credit transfer for MOOCs and give the regulation credit equivalent to the student for the credit earned through a few learning courses.
- Student registration for the MOOCs shall be only through the respective department of the University. It is mandatory for the student to start transfer a minimum of 30 days before.
- Credit transfer policy will be applicable to the Undergraduate (First Semester) student only.
- The respective department shall identify the course pertinent for credit transfer.
- The University shall identify the requirement of the student's learning with existing requirement for the credit transfer.
- The University shall determine the maximum as a Master for credit transfer with the student's requirement of the completion of the credit course.
- The University shall receive an approval of MOOCs transfer with the approval of the University to transfer to the credit of the student. The University will ensure that student must be credit transfer.

will not be granted unless under \$5000, will support the original credit only after successful completion of the course and obtaining a certificate issued by the respective authority along with the proper proof marks and grades.

k) The student shall submit the following to the examination center of the university:

- i) List of students who have passed HSC, entered in the central database along with the signature of concerned.
- ii) A declaration that filed by the students to credit transfer.

l) The student shall submit any documents by which the implementation of the policy from time to time and shall submit its draft to the policy table (list of persons, charges, length of HSC, JNU/VM, NETS, and some provisions).

Note: Student shall be permitted to appear for HSC's without through online platform submitted to the University's database to take.

13. Academic Bank of Credits (ABC)

The University has implemented Academic Bank of Credits (ABC) as follows. According to regulation no. 104/2012:

- a) provide online capability for learners across the network of all the states
- b) provide up to six past the credits through ABC's more appropriate digital platform
- c) set the record of past the credit/degree in the web for academic credit in ABC
- d) support Multiple Entry and Exit system with credit bank, credit transfer and credit recognition from students' record.

14. Mandatory Internships

General Description: The university is introducing after year 10 with all students with a minimum of 60 credit hours. These include one internal and three years, respectively, are mandatory. It shall be completed in collaboration with Local industries, Govt. Organizations, non-profit groups, for employment within ABC's only subject to the strict adherence specifications of the University's program. One of the university's mandatory internship is to be completed in the last year of the program. The student shall be required to submit a certificate of completion with government recognition/ABC's database. The other two steps in the end of the year is industry placement and shall be completed in collaboration with industries. One student shall register for this internship as per curriculum and after completion of year 10, 2016. The guidelines issued by the ABC/ABC University shall be followed for carrying out and evaluation of University's Internship Program and industry knowledge.

Evaluation of the student's internship shall be through the university's committee. It should be required to submit a written statement report in the university's database and upon the set and procedures before the University's committee. The report shall be of the Department, supervisor of the internship and a report form, member of the committee. A statement of record a certificate from industry shall

be available to support the report writing and present to class (up to 15% weight per week). It shall be evaluated for 40% of total marks. There shall be no assessment for the Semester Learning. A student shall score maximum 40% of marks for report writing evaluation. In case, if a student fails, faculty shall suggest to read some research paper on the same topic or conduct the lab activity.

Full Semester Learning and Project work: In the last semester, the students should consistently report and submit learning journal=diary and to provide before the 4th week of a project with half-written abstract on. In the end of the semester the student shall submit an assignment comprising semester end a project report. A student will also be provided to read project report at the next annual conference on learning.

The project is an shall be awarded half-semester credit. The marks to be given project work 30 marks and dissertation shall be 30 marks for journal and 40 marks for abstract submission. The maximum marks the student for 70 marks (Report 30 marks, Journal 30 marks, and for 10% of the semester). If no pass student (no document) of the document for the benefit of discipline and staff and the score is to be calculated by the Department Project Report Committee according to report on a weekly basis and 10% for 70 marks. The annual evaluation of these 70 marks, New VET Program are conducted in the presence of student, an external external committee appointed by the University and is evaluated for 10 marks.

The college will facilitate and provide the student learning program. To facilitate of students a mandatory 1 day student life as a group a schedule, faculty will not be eligible for the award degree. In each year the student shall have a compulsory attendance.

15. Guidelines for offering a Minor

To promote good disciplinary knowledge among the students, the students should take 60 ECTS or a course which should be able to be minor degree. It shall be marked as minor.

1. The Minor program, require the completion of 12 credits in three project classes.
2. The course, faculty shall be able to offer a minor degree to be minor degree. The minor degree, the marks will be the student who have done minor degree, that course. It should be a student, then to study may take a minor degree course. In the past, it is recommended, the student should complete the competency course (3 equivalent) before studying 6.7 the education.
3. It should be marked as 12 credits to complete a total of 12 credits.

Note: A total of 12 ECTS credits is to be offered to the student. A student can complete the requirement for Minor by taking the minor offered through various certificate/degree/program.

16. Guidelines for offering a Minor

The objective of offering 60 ECTS (Minor) is to facilitate the student to choose

References

- H** It is evident from the diagram that the bulk of investment is going to private or at least semi-private institutions – the universities.

Read more about the New Education at

- 8 The rights and interests of students shall apply through the IAC's membership grant structure. The whole process should be completed within six weeks before the start of every semester. The entire students shall be permitted to join the current year in Flowers.
- 9 The student council shall submit their a list to the principal through the local parent department within 10 days. The entire members shall submit the report of student personality to Flowers.
- 10 The students enrolled in the Flowers course will be considered automatically to be admitted to the program. Students who are required to pass a period of study in another institution.
- 11 There will be a 10% reduction of a school fee. Flowers requires officials to be 10% of the entire school members.

Dr. John H. Hamilton III

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12. Presenting Notes

The 1994-1995 study is subject to a number of limitations. In addition to the limitations mentioned previously, a number of

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doi:10.1371/journal.pone.0142262.g002

d The rank scores are ordered by 100 points. The results obtained in each group, as it is essential to a comprehensive study, are given below, ordered as per the steps in which the scores obtained by the student are:

The authors are grateful to Dr. A. F. Kozlovskaya for her help in preparing the manuscript.

Range to add the score to the subject list	Grade	Grade point assigned
70-80	A (Superior)	4
60-70	B (Excellent)	3
50-60	C (Good)	2
40-50	D (Fair)	1
30-40	E (Poor)	0
0-30	F (Fail)	0
Below 0	Not Listed	0

- (i) A student receives credit for a course if the student has completed the course and has received a passing grade for the course (that is, a grade of C or better).
- (ii) For each course, credit earned, "Satisfactory" or "Unsatisfactory," will be determined based on the average grade for the course. If the average grade is satisfactory, the student will receive credit for the course.

For example, if a student's grade point average is 2.0, then the student will receive credit for the course.

The student's grade point average (GPA) is the total number of credit hours earned divided by the total number of credit hours attempted. A student's GPA is calculated by dividing the total number of credit hours earned by the total number of credit hours attempted.

$$\text{GPA} = \frac{\text{Total Credit Hours}}{\text{Total Credit Hours Attempted}}$$

where GPA is the student's grade point average, $\text{Total Credit Hours}$ is the total number of credit hours earned, and $\text{Total Credit Hours Attempted}$ is the total number of credit hours attempted.

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Grade point average (GPA) is a measure of a student's academic performance. It is calculated by dividing the total number of credit hours earned by the total number of credit hours attempted. The student's GPA is calculated by dividing the total number of credit hours earned by the total number of credit hours attempted.

Student GPA

A student's grade point average (GPA) is calculated by dividing the total number of credit hours earned by the total number of credit hours attempted. The student's GPA is calculated by dividing the total number of credit hours earned by the total number of credit hours attempted.

Class (credit)	GPA Score
First Class (1 credit)	2.0
Second Class (2 credit)	2.5
Third Class (3 credit)	3.0
Fourth Class (4 credit)	3.5

$$\text{GPA} = \frac{\text{Total Credit Hours}}{\text{Total Credit Hours Attempted}}$$

18. Work loading of students

If the candidate has any other assignment or laboratory or field work or independent or subjective work assigned from last two years of the candidate will be considered as such work.

19. Multiple Entry / Exit options

19.1 Exit Entry:

The candidate can choose to exit the program in a 3rd and/or 4th semester (table 19.1).

1. **Exit Certificate in Field of study (Bachelor):** Program duration: Five years (three semesters of the undergraduate programme). A candidate is allowed to exit additional to a 36 credit degree course (having two semesters including at least 6 credit) to enter the master's programme that would help the candidate acquire **MA** entry requirements required to enter the master's.

2. **Exit Diploma in Field of study (Bachelor):** Program duration: Four years (three semesters) of the undergraduate programme. A candidate is allowed to exit additional to a 36 credit degree course (having two semesters including at least 6 credit) to enter the master's programme that would help the candidate acquire **MA** entry requirements required to enter the master's.

3. **Exit diploma (Science Faculty of study discipline) i.e., B.Sc. Engineering in Field of study (Bachelor):** Program duration: Four and a half years (three semesters of the undergraduate programme). (Table 19.1)

19.2 Entry Policy:

Students are eligible entry to the candidate that the MA entry requirements will be provided in the course of time.

NOTE: The University authorities will ensure that they agree to the admission of MA entry to exit and that policies have been to ensure that all services that students in the light of policies changes brought by UGC, AICTE and other government.

20. Gap Year Concept

The year concept for student is implemented in a candidate is a school and not working student who wish to pursue into his/her setting / focused intervention for three or more months of one year or any other after 18 years to pursue full time comprehensive programme to work for example. This period may be extended to one year of the first and three (3) years would not be counted for the first time the first entry time for graduation. The principal of the University college that student will be required to attend by the students in the University. The evaluation committee will award by the University that is a year to request (subject to the school and the university does not do it will be given) the student if it is not the first year or not.

15. Technical Regulations

Designs, plans, drawings or related documents are only for the student's personal use. The designs produced after completion of a course registration (Candidate who have been declared for non-attendance or not fulfilled academic requirements) or who have failed after having completed the course or under regulations or have been removed and wish to continue the course are eligible for admission into the curriculum relevant from the date of commencement of that course with the same or equivalent subjects as and where relevant and subject, subject to Section 7 and they must follow the academic regulations that apply to be readmitted.

Candidates who are permitted to study their Year should be eligible for re-enrolling into the following year of study if, from time to time, a decision of admission of other work, will not be Section 7 and they will follow the academic regulations that apply to be readmitted.

16. Maximum instruction days for a semester

The instruction is over one day including lecture and laboratory and field day(s).

17. Division of instruction

The division of instruction of the entire 36 weeks undergraduate programme in Engineering in Pakistan is as following: minimum and maximum semester will be as follows only.

18. Student Exemptions

Student exemption shall be as per the guidelines issued by the Government of Sindh, Pakistan and the University is expected to follow.

19. General Instructions

1. The academic regulations should be read as a whole for purpose of any interpretation.
2. Minor errors (spelling, punctuation) are ignored.
3. Where the words "the", "any", "all", occur in the regulations, they shall signify "the", "any", "all", respectively.
4. In the case of any doubt as to the interpretation of the regulations, the decision of the Vice-Chancellor shall be final.
5. The University may change or amend the academic regulations, or update at any time and the changes or amendments shall be made applicable to all the students as at the date when the change is effected by the University.
6. In the case of any doubt or discrepancy in the interpretation of the regulations from the decision of the Vice-Chancellor / Head of the institution, it shall

ACADEMIC REGULATION (AR)

FOR EIGHTH GRADE ENTRY SCHEME

(Difference for the students admitted from 2021 onwards for all from 1st time directly admitted from 2022-23 onwards)

1. Level of the Degree

- a) Award of **A** Grade Degree / B Grade Degree with a Merit for the full degree following:
 - a1) Pursue a course of study the one that there there additional points and not more than six academic years. However, for the students applying Full-time Entry the period shall be extended by one year of the time **100** days the course would be full due to the maximum period given for the graduate (Six years).
 - a2) Degree for 15+ graduate it requires 1275 credits.
- b) Award of **B** Grade Degree with Honors if there limits the following:
 - b1) Student scores with least 15 credits (3000) as a) the minimum of a B Grade degree is 15 credits.
 - b2) Repeating the degree is allowed.
 - b3) Degree may be completed with Honors with a B Grade program.

2. Students who fail to fulfil the requirement for the level of the degree within six semesters and five years from the year of admission shall obtain the degree.

3. Minimum Academic Requirements

The following academic requirements have to be satisfied by students in the requirements mentioned in every level.

- a. A student shall be deemed to have satisfied the minimum grade requirement to proceed the studies if he/she is each theory, practical, project, drawing subject at present of his/her score not less than 70% of marks in the each examination and a minimum of 60% of marks in the overall of the total examination and and are treated like a degree.
- b. A student shall be deemed to have satisfied the minimum grade requirement to proceed the studies if he/she is each theory, practical, project, drawing subject at present of his/her score not less than 70% of marks in the each examination and a minimum of 60% of marks in the overall of the total examination and and are treated like a degree.

And to meet if student is actually deemed for repeat or even for particular students, the student may apply for the credit through supplementary exam of the above subject for the concentration of 75% and 1 semester class work of year 11th.

4. Conduct Policies

- a. The suspension of a student's academic record is not a punishment.
 - b. A student's rights to appeal the final suspension is a right to be heard, it is not his right to be just and punishment may appear for that reason in the non-punitive/suspensive character of the act.
 - c. When a student is detained due to lack of credit/degree of attendance for studies may be suspension when the suspension is lifted after full work or suspension expires, the student shall be in compliance with all rules before a suspension.
5. All other regulations as may be set by the Board of Trustees, and any degree course of action will be set by the Board of Trustees.

**V23 First Year Course Structure**

I – Semester							
S.No.	Course Code	Category	Name of the Course	L	T	P	C
1	V23B0701	ES&M	Common core English	2	0	0	2
2	V23B0702	ES&M	Engineering Physics	2	0	0	2
3	V23B0703	ES&M	Linear Algebra & Calculus	2	0	0	2
4	V23B0704	Engineering Science	Basic Civil & Mechanical Engineering	1	0	0	1
5	V23B0705	Engineering Science	Introduction to Programming	1	0	0	1
6	V23B0706	ES&M	Common core English Lab	0	0	1	1
7	V23B0707	ES&M	Engineering Physics Lab	0	0	1	1
8	V23B0708	Engineering Science	Engineering Workshop	1	0	1	1.5
9	V23B0709	Engineering Science	Computer Programming Lab	1	0	1	1.5
10	V23B0710	Engineering Science	IT Virtualization	0	0	1	1
11	V23B0701	ES&M	Health and Wellness, Yoga and sports	-	-	1	1.5
Total				14	0	12	20.5

L: Lectures, T: Tutorial, P: Practical, C: Credits

II – Semester							
S.No.	Course Code	Category	Name of the Course	L	T	P	C
1	V23C0701	ES&M	Applied Chemistry	2	0	0	2
2	V23C0702	ES&M	Differential Equations & Vector Calculus	1	0	0	1
3	V23C0703	Engineering Science	Basic Electrical and Electronics Engineering	2	0	0	2
4	V23C0704	Engineering Science	Engineering Graphics	1	0	1	2
5	V23C0705	Professional Core	Data Structures	3	0	0	3
6	V23C0706	ES&M	Applied Chemistry Lab	0	0	2	1
7	V23C0707	Engineering Science	Electrical and Electronics Engineering Workshop	0	0	1	1.5
8	V23C0708	Professional Core	Data Structures Lab	0	0	2	1.5
9	V23C0709	ES&M	III, IV, V, VI, VII, VIII, IX, X, XI, XII, Sports & Games/ Career advice Services	-	-	1	1.5
Total				17	0	17	20.5

L: Lectures, T: Tutorial, P: Practical, C: Credits

B.Tech CSE (Artificial Intelligence)
&
B.Tech.(Artificial Intelligence & Machine Learning)
V23 III Semester Course Structure and Syllabus
Course Structure

III - Semester							
S.No.	Course Code	Category	Courses	L	T	P	C
1.	V23MAT05	B22H	Discrete Mathematics & Graph Theory	3	0	0	3
2.	V23HST03	B22H	Universal human values - Understanding harmony and ethical human conduct	1	1	0	1
3.	V23ACT01	Regulating Interest	Artificial Intelligence	3	0	0	3
4.	V23CST03	Professional Core	Advanced Data Structures & Algorithm Analysis	3	0	0	3
5.	V23CST04	Professional Core	Object Oriented Programming through Java	3	0	0	3
6.	V23CSE04	Professional Core	Advanced Data Structures and Algorithm Analysis Lab	0	0	3	1.5
7.	V23CSE05	Professional Core	Object Oriented Programming Through Java Lab	0	0	3	1.5
8.	V23CSE06	Enhancement Course	Python Programming Lab	0	1	2	1
9.	V23HST09	B22H	Design Thinking & Innovation	1	0	2	1
10.	V23ENT00	MVC	Professional Communication Skills - I	0	3	0	0
total				15	04	10	21

L=Lecture, T=Tutorial, P=Practical, C=Credits

Syllabus

Sem	E-Tech 181 Sem	L	T	P	C	COURSE CODE
Regulation	VTS	3	0	0	3	V23M4T06
Name of the Course	Discrete Mathematics and Graph Theory					
Branch	CSE, CSE(AI), AI & ML, CXT and CSE(CN)					

Syllabus Details

Course Objectives:

- To introduce the students to the topics and techniques of discrete methods and combinatorial reasoning.
- To introduce a wide variety of applications. The algorithmic approach to the solution of problems is fundamental in discrete mathematics, and this approach reinforces the close cut between this discipline and the area of computer science.

At the end of the course, students will be able to:

- Develop skills required for solving mathematical problems using mathematical logic.(LO)
- Demonstrate the set theory principles, relations and functions in real-time situations.(O)
- Apply the knowledge of combinatorics and recursive relations in formulating and solving complex problems.(O)
- Apply the graph theory principles and techniques in computer science-related problems.(LO)
- Find shortest paths and minimal spanning trees using Prim's and Kruskal's Algorithms, BFS and DFS algorithms.(CO)

UNIT-I: Mathematical Logic

Propositional Calculus: Statements and Associates, Connectives, Well Formed Formulas, Truth Tables, Tautologies, Equivalences of Formulas, Duality Law, Tautological Implications, Normal Forms, Theory of Inference for Statement Calculus, Consistency of formulas, indirect method of Proof, Predicate Calculus: Predicate, Predicate Logic, Quantifier Functions, variables and Quantifiers, Free and Bound Variables, Inference Theory for Predicate Calculus.

UNIT-II: Set theory

Sets, Operations on Sets, Principle of Inclusion-Exclusion, Relations: Properties, Equivalence, Partition and Covering, Transitive Closure, Equivalence, Compatibility and Partial Ordering, Hasse Diagrams, Functions: Bijective, Composite, Inverse, Permutation,

and Recurrence Relations, Limits and its Properties.

UNIT-IV: Combinatorics and Recurrence Relations

Basic of Counting, Permutations, Permutations with Repetitions, Circular and Restricted Permutations, Combinations, Restricted Combinations, Binomial and Multinomial Coefficients and Theorems.

Recurrence Relations:

Generating Functions, Functions of Sequences, Partial Fractions, Calculating Coefficients of Generating Functions, Recurrence Relations, Formulation in Recurrence Relations, Solving Recurrence Relations by Substitution and Generating Functions, Method of Characteristic Roots, Solving Homogeneous and Recurrence Relations.

UNIT-V: Graph Theory

Basic Concepts, Graph Theory and its Applications, Sub graphs, Graph Representations: Adjacency and Incidence Matrices, Isomorphic Graphs, Paths and Circuits, Eulerian and Hamiltonian Graphs.

UNIT-VI: Multi graphs

Multi graphs, Bipartite and Planar Graphs, Euler's Theorem, Graph Colouring and Covering, Chromatic Number, Spanning Trees, Prim's and Kruskal's Algorithms, DFS and BFS Spanning Trees.

TEXTBOOKS:

1. Discrete Mathematical Structures with Applications to Computer Science, J.F.Treadaway and P. Handan, Tata McGraw Hill.
2. Elements of Discrete Mathematics- A Computer Oriented Approach, C.L.Luo and Q. F. Mahapatra, 3rd Edition, Tata Mc Graw Hill.
3. Theory and Problems of Discrete Mathematics, S. K. Ghosh's, S. K. Ghosh, Seymour Lipschutz and Marc Lutz Lipson, 3rd Edition, McGraw Hill.

REFERENCE BOOKS:

1. Discrete Mathematics for Computer Scientists and Mathematicians, J.L. Hoot, A. Kandel and T. F. Baker, 2nd Edition, Prentice Hall of India.
2. Discrete Mathematical Structures, Theory and Exercises, Robert C. Walby, 1st Edition, Coffer Press, 1981.
3. Discrete Mathematics, S.K. Chakrabarty and B.K. Datta, Oxford, 2011.
4. Discrete Mathematics and its Applications with Combinatorics and Graph Theory, K.H.Niven, 7th Edition, Tata Mc Graw Hill.

Year/Sem	II II Tech I Sem	L	T	P	C	COURSE CODE
Regulation	V23	2	1	0	1	V23MMT23
Name of the Course	Universal Human Values – Understanding Harmony and ethical human conduct					
Branch	CSE,CSE(AI), AI & ML, CST, and CSE(DS)					

Syllabus Details

Course Objectives

- To help the students appreciate the essential complementarity between *Valued* and *being* to ensure sustained happiness and prosperity which are the core aspirations of all human beings.
- To facilitate the development of 4 holistic perspectives among students towards life and profession as well as towards happiness and prosperity based on a correct understanding of the Human reality and the rest of existence. Such holistic perspective forms the basis of Universal Human Values and movement towards value-based living a natural way.
- To highlight plausible implications of such a Holistic understanding in terms of ethical human conduct, fruitful and mutually fulfilling human behaviour and mutually searching interaction with nature.

Course Outcomes

- Define the terms like Natural Acceptance, Happiness and Prosperity.(1.1,1.2)
- Identify one's self and one's surroundings (family, society, nature).(1.1,1.2)
- Apply what they have learnt to their own self and their day-to-day settings in real life (1.2)
- Relate human values with human relationship and human society (1.4)
- Justify the need for universal human values and harmonious existence(1.4)
- Develop as socially and ecologically responsible engineers(1.5,1.6)

Course Topics

The course has 28 lectures and 14 tutorials in 8 modules. The lectures and tutorials are of 1-hour duration. Tutorial sessions are to be used to explore and practice what has been proposed during the lecture sessions. The Teacher's Manual provides the outline for lectures as well as practicum sessions. The teacher is expected to present the issues to be discussed as propositions and encourage the students to have a dialogue.

- UNIT I** Introduction to Value Education
(6 lectures and 3 tutorials for practice session)
Lecture 1: Right Understanding, Relationship and Physical Facility (Bhakti movement and the Role of Education)
Lecture 2: Understanding Value Education
Tutorial 1: Practice Session P11 Sharing about oneself
Lecture 3: Self-exploration as the Process for Value Education
Lecture 4: Contentment, Happiness and Prosperity – the Basic Human Aspirations
Tutorial 2: Practice Session P12 Exploring Human Consciousness
Lecture 5: Happiness and Prosperity – Current Systems
Lecture 6: Method to Fulfil the Basic Human Aspirations
Tutorial 3: Practice Session P13 Exploring Natural Acceptance
- UNIT II** Harmony in the Human Being
(6 lectures and 3 tutorials for practice session)
Lecture 7: Understanding Human being as the co-existence of the self and the body
Lecture 8: Distinguishing between the Seeds of the self and the body
Tutorial 4: Practice Session P14 Exploring the difference of seeds of self and body
Lecture 9: The body as an instrument of the self
Lecture 10: Understanding Harmony in the self
Tutorial 5: Practice Session P15 Exploring Sources of imagination in the self
Lecture 11: Harmony of the self with the body
Lecture 12: Programmes to ensure self-regulation and health
Tutorial 6: Practice Session P16 Exploring Harmony of self with the body
- UNIT III** Harmony in the Family and Society
(6 lectures and 4 tutorials for practice session)
Lecture 13: Harmony in the Family – the Basic Unit of Human Interaction
Lecture 14: 'Trust' – the Foundational Value in Relationship
Tutorial 7: Practice Session P17 Exploring the Feeling of Trust
Lecture 15: 'Respect' – as the Right Evaluation
Tutorial 8: Practice Session P18 Exploring the Feeling of Respect
Lecture 16: Other feelings, justice in human-to-human relationship
Lecture 17: Understanding Harmony in the Society
Lecture 18: Vision for the Universal Human Order

Tutorial & Practice Session P18 Exploring Systems to fulfill Human Goal

- UNIT IV** Harmony in the Nature / Existence
(4 lectures and 2 tutorials for practice session)
Lecture 17: Understanding Harmony in the Nature
nature for interdependence, self-sufficiency and mutual fulfillment
among the Four Orders of Nature
Tutorial 18: Practice Session P18 Exploring the Four Orders of Nature
Lecture 21: Realizing Existence as Co-existence at all Levels
Lecture 22: The Holistic Perception of Harmony in Existence
Tutorial 23: Practice Session P23 Exploring Co-existence in Existence
- UNIT V** Implications of the Holistic Understanding – a path to Holistic Well-Being
Ethics (6 lectures and 2 tutorials for practice session)
Lecture 23: Natural Acceptance of Human Values Lecture 24:
the Universality of (Ethical) Human Conduct
Tutorial 25: Practice Session P25 Exploring ethical Human Conduct
Lecture 25: A Basis for Humanistic Education, Humanistic Contribution and
Universal Human Order
Lecture 26: Competence in Professional Ethics
Tutorial 26: Practice Session P26 Exploring Humanistic Studies in
Education Lecture 27: Holistic Technologies, Production Systems and
Management Models-Typical Case Studies
Lecture 28: Strategies for Transition towards Value-based Life and
Profession Tutorial 28 Practice Session P28 Exploring Steps of Transition
towards Universal Human Order

Practice Sessions for UNIT I – Introduction to Value Education

- P11 Exploring about oneself
- P12 Exploring Human Consciousness
- P13 Exploring Natural Acceptance

Practice Sessions for UNIT II – Harmony in the Human Being

- P14 Exploring the difference of Needs of self and body
- P15 Exploring Sources of Inspiration in the self
- P16 Exploring Harmony of self with the body

Practice Sessions for UNIT III – Harmony in the Family and Society

- P17 Exploring the Feeling of Trust
- P18 Exploring the Feeling of Respect

WE Exploring Mysteries in LIFE human body

Practice Sessions for UNIT IV – Harmony in the Nature (Existence)

PS10 Exploring the Four Orders of Nature

PS11 Exploring Co-existence in Nature

Practice Sessions for UNIT V–Implications of the Values Understandings Look at Professional Ethics

PS12 Exploring Ethical Human Conduct

PS13 Exploring Fundamental Models in Education

PS14 Exploring Steps of Transition towards Universal human Order

READING:

Textbook and Teachers Manual

• The Textbook:

R. K. Gaur, R. Arshiana, C. P. Sagar, *A Foundation Course in Human Values and Professional Ethics*, 2nd Revised Edition, Excel Books, New Delhi, 2019, ISBN 978-93-87334-47-1

• The Teacher's Manual

R. K. Gaur, R. Arshiana, C. P. Sagar, *Teachers' Manual for A Foundation Course in Human Values and Professional Ethics*, 2nd Revised Edition, Excel Books, New Delhi, 2019, ISBN 978-93-87334-55-2

Reference Books

1. *Swatantra Indhiya*, B. K. Chakravarty, C. P. Sagar, Jaipur, Vikas Publications, 2019, ISBN 978-93-87334-55-2
2. *Swatantra Indhiya*, B. K. Chakravarty, New Age Text Publishers, New Delhi, 2019.
3. *The Story of Dog* (Book)
4. *The Story of My Experiments with Truth* by Mahatma Jyotirao Phule
5. *Small is beautiful* E. F. Schumacher
6. *Stone is Beautiful*– Carlo Andrucci
7. *Economy of Farmhouse* J. C. Karmacharya
8. *Global story Against Big-Panda* Tushar K.
9. *Rediscovering India* by Shriharshal
10. *A Wind Swept in Indian Home Rule* by Mahatma K. Gandhi
11. *India 1947* Panchanabandana Kishor Kishor Das
12. *Vishwakarma-Krishna Kalland* (English)
13. *Gandhi Karmacharya Kalland* (English)

use of lecture:

Lecture hours are to be used for interactive discussion, placing the proposals about the topics at hand and motivating students to reflect, analyse and verify them.

Tutorial hours are to be used for practice sessions.

While analysing and discussing the topic, the faculty mentor's role is in pointing to essential elements to help in sorting them out from the surface elements. In other words, help the students explore the important or critical elements.

In the discussions, particularly during practice sessions (tutorial's), the mentor encourages the student to connect with one's own self and do self-observation, self-reflection and self-exploration.

Students may be asked to submit diaries. The student is encouraged to take up "ordinary" situations rather than "extra-ordinary" situations. Such observations and their analyses are shared and discussed with other students and faculty mentor, in a group setting.

Tutorial's (experiments or practical's) are important for the course. The difference is that the laboratory is everyday life, and practical are how you behave and work in real life. Depending on the nature of topics, worksheets, home assignments and/or activity are included.

Semester	III	L	T	F	C	COURSE CODE
Regulation	VT2	3	0	0	3	VT2A1TC1
Name of the Course	Artificial Intelligence					
Branch	Common to CSE(A) and AISEPL					

Syllabus Details

Course Outcome:

After Successful completion of the Course, the student will be able to:

- CO1: Discuss the foundational concepts of AI. (K2)
- CO2: Demonstrate search strategies for problem solving. (K4)
- CO3: Illustrate Knowledge Representation approaches. (K3)
- CO4: Demonstrate Reasoning approaches. (K3)
- CO5: Illustrate architecture and role of Expert Systems. (K3)

UNIT - I

Introduction: AI problems, foundation of AI and history of AI, intelligent agents: Agents and Environments, the concept of rationality, the nature of environments, structure of agents, problem solving agents, problem formulation.

UNIT - II

Searching: Searching for solutions, uniformed search strategies - Breadth first search, depth first Search, Search with partial information (Heuristic search), Hill climbing, A*, AO* Algorithms, Problem reduction, Game Playing-Adversarial search, Minimax algorithm, optimal decision in multiplayer games, Problem in Game playing, Alpha-Beta pruning, Evaluation functions

UNIT - III

Representation of Knowledge: Knowledge representation - propos, predicate logic- logic programming, semantic nets- frames and inheritance, constraint propagation, representing knowledge using rules, rules based deduction systems, Reasoning under uncertainty, review of probability, Bayes' probabilistic inference and Bayesian belief theory.

UNIT - IV

Logic concepts: First order logic, Inference in First order logic, Propositional vs. First Order inference, Deduction & Lifting, Forward chaining, Backward chaining, Resolution, Learning from observations Inductive learning, Decision trees, Expertise based learning, Statistical Learning methods: Reinforcement Learning.

UNIT - V

Expert Systems: Architecture of expert systems, Rules of expert systems - Knowledge Acquisition, Meta knowledge, Heuristics, Typical expert systems - MYCIN, DART, XCON: expert system shells.

Textbooks:

1. Artificial Intelligence - A Modern Approach, T. Russell and P. Norvig, Third Edition, Pearson Education.
2. Artificial Intelligence, Elaine Rich and Kevin Knight, Global Edition, Tata Mc Graw Hill.

Reference Works:

1. Computational Intelligence: a logical approach, David Poole, Allen Mackworth, Randy Goebel, Oxford University Press.
2. Artificial Intelligence: Structures and Strategies for complex problem solving, G.Luger, Fourth Edition, Pearson Education.
3. Artificial Intelligence: A new Synthesis, J. Nilsson, Elsevier Publishers.
4. Artificial Intelligence, Arno Kruschke, Springer Learning.

On Line Learning Resources:

1. <https://www.google/>
2. <https://www.programs.com/edu/soft/ai/ai-program>

Semester	88	L	T	P	C	COURSE CODE
Regulation	VT2	3	2	2	2	VTSC0704
Name of the Course	Advanced Data Structures & Algorithms Analysis					
Branch	Common to CSE, CSE, CSE(AI), AIESEC and CSE (SV)					

Syllabus Details

Course Outcomes

After successful completion of the course, the student will be able to:

CO1: Demonstrate asymptotic notation and non-linear data structures like AVL and B-Trees.(K1)

CO2: Demonstrate graphs and divide and conquer techniques (K1)

CO3: Use greedy approach to determine various problems (K1)

CO4: Apply Dynamic Programming and Backtracking techniques (K1)

CO5: Solve different problems using Branch and Bound, NP Hard and NP Complete (K1)

UNIT-I

Introduction to algorithm Analysis, Space and Time Complexity analysis, Asymptotic Notations

AVL Trees - Creation, Insertion, Deletion operations and Applications

B-Trees- Creation, Insertion, Deletion operations and Applications

UNIT-II

Heap Trees (Priority Queues) - Min and Max Heaps, Operations and Applications

Graphs - Terminology, Representations, Basic search and Traversal, Connected Components and Disconnected Components, applications

Brute and Naïve: The naive method, quick sort, merge sort, Strassen's matrix multiplication, Towers of Hanoi

UNIT-III

Greedy Methods: General Method, Job Sequencing with deadlines, Knapsack Problem, Minimum cost spanning trees, Single Source Shortest Path- Dijkstra's

UNIT-IV

Dynamic Programming: General Method, All pairs shortest paths, Single Source Shortest Path-General Weight (Bellman Ford Algorithm), Dynamic Binary search trees, 0/1 Knapsack, String Editing, Travelling Salesman problem.

Backtracking: naive method, N-Queens Problem, Sum of subsets problem, Graph Coloring, 0/1 Knapsack Problem.

UNIT-V

Branch and Bound: The General Method, 0/1 Knapsack Problem, Travelling Salesperson problem

NP Hard and NP Complete Problems: Basic Concepts, Cook's theorem (without Proof)

NP Hard Graph Problems: Clique Decision Problem (CSP), Travelling Salesperson Decision Problem (TSP).

NP Hard Scheduling Problems: Job Shop Scheduling.

Textbooks

1. Fundamentals of Data Structures Horowitz, Ellis, Sahni, Samsi, Mahla, Dhawan, 2nd Edition Universities Press

Reference Books

1. Data Structures and program design in C, Robert Kruse, Pearson Education Asia
2. An introduction to Data Structures with applications, Tremblay & Sorenson, McGraw Hill
3. The Art of Computer Programming, Vol.1: Fundamental Algorithms, Donald E Knuth, Addison-Wesley, 1977
4. Data Structures using C & C++ Language, Anurag K.Tanwar, Pearson, 1996
5. Algorithms + Data Structures + Programs, A. Aho, M. M.
6. Fundamentals of Data Structures in C++, Horowitz Sahni & Mahla, Galgotia Pub.
7. Data Structures in java, Thomas standish, Pearson education asia

Online Learning Resources

1. http://www.tutorialspoint.com/advanced_data_structures/index.asp
2. <http://pete-nelson.net/algorithms.html>
3. Alid al Bari, Introduction to Algorithms (youtube.com)

Semester	III	L	T	P	E	COURSE CODE
Regulation	173	1	0	0	1	VB000018
Name of the course	Object Oriented Programming Through java					
Search	COURSE IN C++, C#, C#, J#, J#, J#, J# AND C# (J#)					

Syllabus Details

Course Outcomes:

After successful completion of the Course, the student will be able to:
 CO1: Demonstrate the object-oriented programming principles with java programming environment. [K1]

CO2: Demonstrate the concepts like classes, objects, argument passing mechanism, overloading and overriding. [K2]

CO3: Illustrate the concepts of arrays, inheritance and interfaces. [K3]

CO4: Demonstrate packages, java libraries, exception handling (java I/O and File handling). [K4]

CO5: Illustrate the concepts of string handling, multithreading and java FX GUI. [K5]

UNIT-I

Object Oriented Programming: Basic concepts, Principles, Program Structure in java: Introduction, Writing Simple java Programs, Mainclass or Package in java Programs, java Statements, Command Line Arguments, User Input in Programs, Simple Sequential Concepts, Programming style.

Data Types, Variables, and Operators: Introduction, Data Types in java, Declaration of Variables, Data Types, Type Casting, Scope of Variable Identifier, Literal Constants, Symbolic Constants, Formatted Output with printf() Method, Static Variables and Methods, Attributes Final, Introduction to Operators, Precedence and Associativity of operators, Assignment operator (=), Basic arithmetic operators, Increment (++ and decrement (--)) operators, Unary operators, Relational operators, Boolean Logical Operators, Bitwise Logical Operators.

Control Statements: Introduction, if Expression, Nested if Expressions, if-else Expressions, Ternary Operator?, Switch Statement, Decision Statements, while Expression, do-while Loop, for Loop, Nested for Loop, For-Each for Loop, Break Statement, Continue Statement.

UNIT-II

Classes and Objects: Introduction, Class Declaration and Modifiers, Class Members, Declaration of Class Objects, Assigning One Object to Another, Access Control for Class Members, Accessing Private Members of Class, Constructor Methods for Class, Overloaded Constructor Methods, Nested Classes, Final Class and Methods, Passing Arguments by Value and by Reference, Keyword this.

Methods: Introduction, Defining Methods, overloaded methods, overloaded constructor Methods, Class Objects as Parameters in Methods, Access Control, Recursive Methods, Naming of Methods, Overriding Methods, Attributes Final and Static.

UNIT-48

Arrays: Introduction, Declaration and Initialization of Arrays, Storage of Array in Computer Memory, Accessing Elements of Arrays, Operations on array Elements, Assigning Array to Another Array, Dynamic Change of Array Size, Sorting of Arrays, Search for Values in Arrays, Class Arrays, Two-dimensional Arrays, Arrays of Varying Lengths, Three-dimensional Arrays, Arrays as Vectors.

Inheritance: Introduction, Process of Inheritance, Types of Inheritance, Unusual Super class-object class, Relinquishing Inheritance of class using final, Access Control and Inheritance, Multilevel Inheritance, Application of Keyword Super, Constructor Method and Inheritance, Method Overriding, Dynamic Method Dispatch, Abstract Classes, Interfaces and Inheritance.

Interfaces: Introduction, Declaration of Interface, Implementation of Interface, Multiple interfaces, Nested interfaces, Inheritance of interfaces, Default methods in interfaces, Static Methods in interfaces, Functional interfaces, Annotations.

UNIT-49

Packages and Java Library: Introduction, Defining Package, Importing Packages and Classes into Programs, Path and Class Path, Access Control, Packages in Java SE, Invoking Packages and its classes, Class Object, Enumeration, class Path, Wrapper classes, Auto-boxing and Auto-unboxing, java util classes and interfaces, Formatter class, Main class, Time Package, Date classes (java.time.*), Formatting for Date/Time in java, Temporal Adjuster Class, Temporal Adjuster Class.

Exception Handling: Introduction, Hierarchy of Standard Exception Classes, Represents errors and throws, try, catch, and finally blocks, Multiple Catch Clauses, Class Throwable, Unchecked Exceptions, Checked Exceptions.

Java I/O and File: Java I/O API, standard I/O streams, types, Byte streams, Character streams, Reader class, Writer class, java.io.* (Text Book 2)

UNIT-50

String Handling in Java: Introduction, Interface Char Sequence, Class String, Methods for Extracting Characters from Strings, Comparison, Modifying, Searching, Class StringBuffer.

Multi-threaded Programming: Introduction, Need for Multiple Threads, Multi-threaded Programming for multi-core Processors, Thread Class, Main Thread-Creation of new Threads, Thread States, Thread Priority-Synchronization, Deadlock and Race Conditions, Inter-thread Communication - Suspending, Resuming and Stopping of Threads.

Java FX GUI: Java FX Scene Builder, Java FX App Window Structure, displaying text and image, event handling, laying out nodes in scene graph, mouse events (Text Book 3)

Textbook

1. JAVA one step ahead, Anitha Sethi, B.L.S. publishers, Oxford.
2. Jey web (JAVA, Fundamentals of Object Oriented Programming, Debasis Senapati, Murali & Sons, Cambridge, 2021.
3. JAVA 8 for Programmers, Paul Dais, Harvey Mudd, 4th Edition, Pearson.

Reference Books:

1. The complete Reference Java, Lathiadas, Markert Scholdt, TMH.

1. Introduction to Java programming, 7th edition, © Paul D. Song, Pearson

Online Learning Resources:

1. <http://wpi.ac.in/courses/106/106/206106191/>
2. http://wpi.ac.in/guided-learning/206/106/206106191/206106191_47618614617_shared/courses/

Semester	8th	L	T	F	C	COURSE CODE
Regulation	CT8	5	5	5	1.5	VT8EC801
Name of the course	Advanced Data Structures & Algorithms Analysis I-2					
Branch	Common to CSE, CTE, CBE(AI), CBEML and CBE (DE)					

Syllabus Details

Course Outcomes

After Successful completion of the Course, the student will be able to:

CO1: Demonstrate program on AVL tree and B+ tree. (K1)

CO2: Develop programs on Sorting algorithms and Graph traversal algorithms. (K2)

CO3: Develop programs using Greedy and Dynamic programming technique. (K3)

CO4: Develop programs using Backtracking and branch & Bound technique. (K4)

Experiments covering the Topics:

- Operations on AVL tree, B+ tree
- Graph Traversal
- Sorting techniques
- Minimum cost spanning trees
- Shortest path algorithms
- 0/1 Knapsack Problem
- Traveling Salesman problem
- Optimal Binary Search Tree
- N-Queens Problem
- Job Sequencing

Sample Programs:

1. Construct an AVL tree for a given set of elements which are stored in a file and implement insert and delete operation on the constructed tree. Write contents of tree into a new file using pre-order.
2. Construct B-Trees an order of 5 with a set of 100 random elements stored in array. Implement searching, insertion and deletion operations.
3. Construct min and Max heap using array, delete any element and display the content of the heap.
4. Demonstrate DFT and IDFT for given graph, when graph is represented by:
 - a) Adjacency Matrix b) Adjacency Lists
5. Develop a program for finding the hamiltonian components in a given graph.
6. Implement Quick sort and Merge sort and observe the execution time for various input sizes(Average, Worst and Best cases)
7. Measure the performance of single source shortest path using greedy method

when the graph is represented by adjacency matrix and adjacency lists.

6. Demonstrate Job Scheduling with deadlines using Greedy strategy.
7. Demonstrate a program to solve 0/1 Knapsack problem using Dynamic Programming.
8. Demonstrate N-Queens Problem using Backtracking.
9. Use backtracking strategy to solve n/n mapcolor problem.
10. Demonstrate Travelling Sales Person problem using Branch and Bound approach.

Reference Books:

1. Fundamentals of Data Structures, Horowitz Elia, Sahni Sarnaj, Mahla, Omesh, 2nd edition, University Press.
2. Computer Algorithms C++ Elia Horowitz, Sarnaj, Sahni, Sangchavar, Saunders, 2nd edition, University Press.
3. Data Structures and program design in C, Robert Kruse, Pearson Education Asia.
4. An introduction to Data Structures with applications, Franklin A. Gutwin, McGraw Hill.

Online Learning Resources:

1. <http://cs456-06b.stanford.in/>
2. <http://patheisha.net/Algorithms.html>

Semester	III	L	T	P	C	COURSE CODE
Regulation	V23	0	0	2	1.5	V23CE181
Name of the Course	Object Oriented Programming Through Java Lab					
Prerequisites	Common to CIL, ISE, CSE&IT, BBA&B and EIL (IT)					

3.1. Lab details

Course Outcomes

After successful completion of the Course, the student will be able to:

- CO1 Construct programs to handle classes and objects. (K1)
- CO2 Develop programs that incorporate inheritance and interfaces. (K2)
- CO3 Construct programs on exception handling and file I/O. (K3)
- CO4 Develop programs using multithreading and Java FX. (K4)

Experiments covering the Topics

- Object Oriented Programming fundamentals: data types, control structures
- classes, methods, objects, inheritance, polymorphism,
- Exception handling, Threads, Packages, Interfaces
- Files, I/O streams, Java FX GUI.

Exercise - 1:

- a) Develop a JAVA program to display default value of all primitive data type of JAVA.
- b) Develop a java program that display the roots of a quadratic equation: ax²+bx+c=0 find the discriminant D and based on value of D, describe the nature of root.

Exercise - 2:

- a) Develop a JAVA program to search for an element in a given list of elements using binary search mechanism.
- b) Develop a JAVA program to sort for an element in a given list of elements using bubble sort.
- c) Develop a java program using String Buffer to follow, reversing character.

Exercise - 3:

- a) Develop a JAVA program to implement class Mechanism. Build a class, methods and invoke them inside main method.
- b) Develop a java program to implement method overloading.
- c) Develop a JAVA program to implement constructor.
- d) Develop a java program to implement constructor overloading.

Exercise - 4:

- a) Develop a JAVA program to implement Single inheritance.
- b) Develop a JAVA program to implement multi-level inheritance.
- c) Develop a JAVA program for abstract class to find areas of different shapes.

Exercise – R

- Develop a J2SE program on 'super' keyword.
- Develop a J2SE program to implement Interface. What kind of Inheritance can be achieved?
- Develop a J2SE program that implements Runtime polymorphism.

Exercise – R

- Develop a J2SE program that describes exception handling mechanism.
- Develop a J2SE program on Multi's catch clauses.
- Develop a J2SE program to generate Built-in Exceptions.
- Develop a J2SE program to generate User Defined Exceptions.

Exercise – T

- Develop a J2SE program that creates threads by extending Thread class. First thread display 'Good morning' every 1 sec, the second thread displays 'Hello' every 2 seconds and the third display 'Welcome' every 3 seconds (Repeat the same by implementing Runnable).
- Develop a program on `wait()` and `join()`.
- Demonstrate Deadlock Thread.
- Demonstrate Producer Consumer Problem.

Exercise – R

- Develop a J2SE program that import and use the user defined packages.
- Demonstrate a GUI that display text in label and image in an ImageView (use `Graphics`) in their writing any code.
- Develop a J2SE Calculator app using several `Graphics` components and learn how to respond to user interaction with the GUI.

Textbook

- J2SE one step ahead, Anthea Park, R.T. Jeyapala, Oxford.
- Jay with JAVA, Fundamentals of Object Oriented Programming, Debasmita Ghanta, Second edition, Cambridge, 2018.
- JAVA for Programmers, Paul Dietel, Harvey Deitel, 4th Edition, Pearson.

Reference website

- The complete Reference Java, 11th edition, Herbert Schildt, TMH.
- Introduction to Java programming, 7th Edition, Y Daniel Liang, Pearson.

Online Learning Resources

- <https://github.com/courses/108/303/306003191/>
- https://info.purpleguard.com/blog/en/app/ios/ios_such_0120804642_47618216047_shared/overview

Semester	88	L	T	F	C	COURSE CODE
Regulation	2019	6	1	2	2	VT88C00005
Name of the course	Python Programming Lab					
Branch	Computer Science, CSE, CSE(AI), CSE(CS), CSE(IT) and EEE					

Syllabus Details

Course Outcomes

- After Successful completion of the Course, the student will be able to
- CO1: Illustrate basic concepts and control structures in python programming. (K1)
 - CO2: Demonstrate functions and packages. (K2)
 - CO3: Construct Python programs using structured data types. (K2)
 - CO4: Develop programs on File, Exception handling and OOPs Concepts. (K2)
 - CO5: Construct programs for Data Analysis using Num Py and Pandas. (K3)

UNIT-1

History of Python Programming Language, Three States of Python, Installing Anaconda, Python Distributions, Installing and Using Jupyter Notebook

Parts of Python Programming Language: Identifiers, Keywords, Statements and Expressions, Variables, Operators, Precedence and Associativity, Data Types, Indentation, Comments, Reading Input, Print Output, Type Conversions, the type () Function and is Operator, Dynamic and Strongly Typed Language.

Control Flow Statements: if statement, if-else statement, if-elif-else, Nested if statement, while Loop, for Loop, continue and break Statements, Catching Exceptions Using try and except Statement.

Sample experiments:

1. Develop a program to find the largest element among three numbers.
2. Develop a Program to display all prime numbers within an interval.
3. Develop a program to swap two numbers without using a temporary variable.
4. Demonstrate the following Operators in Python with suitable examples.
 - (i) Arithmetic Operators (ii) Relational Operators (iii) Assignment Operators (iv) Logical Operators
 - (v) Bit-wise Operators (vi) Tertiary operator (vii) Membership operators (viii) Identity Operators
5. Develop a program to add and multiply complex numbers.
6. Develop a program to print multiplication table of a given number.

UNIT-2

Functions: Multi-line Functions, functionally used Modules, Function Definition and Calling the function, return statement and void function, Scope and Lifetime of Variables, Default parameters, Keyword arguments, *args and **kwargs, function time signatures.

Strings: Creating and Sharing Strings, Basic String Operations, Accessing Characters in String by Index Number, String Slicing and joining, String Methods, Formatting Strings.

String: Creating Lists, Basic List Operations, Indexing and Slicing in Lists, Built-in Functions Used on Lists, List Methods, del statement

Sample Experiments:

1. Develop a program to define a function with multiple return values
2. Develop a program to define a function using default arguments
3. Develop a program to find the length of the string without using any library functions.
4. Develop a program to check if the substring is present in a given string or not.
5. Develop a program to perform the given operations on a list:
(i) addition (ii) insertion (iii) slicing
6. Develop a program to perform any 5 to 10 functions by using any list

UNIT-03

Dictionary: Creating Dictionary, Accessing and Modifying Key-Value Pairs in Dictionaries, Built-in Functions Used on Dictionaries, Dictionary Methods, del statement, Tuples and Set: Creating Tuples, Basic Tuple Operations, tuple() Function, Indexing and Slicing in Tuples, Built-in Functions Used on Tuples, Relation between Tuple and List, Relation between Tuple and Dictionary, Using zip() Function, len, len Methods, len() Function

Sample Experiments:

1. Develop a program to create tuples (name, age, address, college) for at least two members and concatenate the tuples and print the concatenated tuples.
2. Develop a program to count the number of vowels in a string (No control flow allowed).
3. Develop a program to check if a given key exists in a dictionary or not.
4. Develop a program to add a new key-value pair to an existing dictionary.
5. Develop a program to sum all the items in a given dictionary.

UNIT-04

File: Types of Files, Creating and Reading Text Data, File Methods to Read and Write Data, Reading and Writing Binary Files, pickle Module, Reading and Writing CSV Files, Python os and os.path Modules

Object-Oriented Programming: Classes and Objects, Creating Classes in Python, Creating Objects in Python, Constructor Method, Classes with Multiple Objects, Class Attributes (vs Data Attributes), Encapsulation, Inheritance, Polymorphism

Sample Experiments:

1. Develop a program to sort words in a file and put them in another file. The output file should have only lower-case words, so any upper-case words from source must be lowered.
2. Develop a Python program to print each line of a file in reverse order.

- Develop a Python program to compute the number of characters, words and lines in a file.
- Develop a program to create, display, append, insert and reverse the order of the items in the array.
- Develop a program to add, transpose and multiply two matrices.
- Develop a Python program to create a class that represents a shape. Include methods to calculate its area and perimeter. Implement subclasses for different shapes like circle, triangle, and square.

UNIT-V

Introduction to Data Science: Functional Programming (DOM and DOM in Python, NumPy with Python, Pandas).

Sample Experiments:

- 1. Develop a Python program to check whether a (100) string contains complex set or not.
- 2. Demonstrate NumPy Arrays creation using `array()` function.
- 3. Demonstrate use of `ndim`, `shape`, `size`, `dtype`.
- 4. Demonstrate basic string, integer and Boolean indexing.
- 5. Develop a Python program to find min, max, sum, cumulative sum of array.
- 6. Prepare a dictionary with at least five keys and each key represent value as a list where the list contains at least ten values and convert this dictionary as a pandas data frame and explore the data through the data frame as follows:
 - a) `apply head()` function to the pandas data frame
 - b) Perform various data selection operations on Data Frame
- 7. Select any two columns from the above data frame, and observe the change in one attribute with respect to other attribute with scatter and plot operations in `matplotlib`.

Reference Books:

- 1. Gaur Shankar S, Vasez S., Introduction to Python Programming, CRC Press.
- 2. Python Programming, S Sridhar, J Jeyamathi, V M Manikaran, 2nd Edition, Pearson, 2024.
- 3. Introduction to Programming Using Python, Y. Daniel Liang, Pearson.

Online Learning Resources:

- 1. <https://www.coursera.org/learn/python-for-applied-data-science-ai>
- 2. <https://www.coursera.org/learn/python-application-pythonsyllabus>

Semester	88	L	T	F	C	COURSE CODE
Regulation	VT2	1	2	2	2	VT2DET09
Name of the Course	Design Thinking & Innovation					
Branch	Computer to Electronics					

Syllabus Details

Course Outcomes

After successful completion of the course, the student will be able to:

- CO1: Define the concepts related to design thinking (CT)
- CO2: Explain the fundamentals of Design Thinking and innovation (CT)
- CO3: Apply the design thinking techniques for solving problems in various sectors (CT)
- CO4: Analyse to work in a multidisciplinary environment (CT)
- CO5: Evaluate the value of creativity (CT)

UNIT-1

Introduction to Design Thinking

Introduction to elements and principles of Design, basics of design-dot line, shape, form, as fundamental design components. Principles of design, Introduction to design thinking, history of Design Thinking, New materials in industry.

UNIT-2

Design Thinking Process

Design thinking process (empathize, analyze, ideate, prototype), implementing the process in driving innovations, design thinking in social innovations, Tools of design thinking- person, customer, journey map, brainstorming, product development.

Activity: Every student presents their idea in three minutes. Every student can present design process in the form of flow diagram or flow chart etc. Every student should explain about product development.

UNIT-3

Innovation

Art of innovation, Difference between invention and creativity, role of creativity and innovation in organizations, Creativity to innovation, Factors for innovation, Measuring the impact and value of innovation.

Activity: Debate on invention and creativity, View and planning from idea to innovation, Debate on value-based innovation.

UNIT-4

Product Design

Problem formulation, introduction to product design, Product strategy, Product value, Product planning, product specification, innovation towards product design case studies.

Activity: Importance of modeling, how to set specifications, Explaining their own product design.

Unit 2-4

Design Thinking in Business Processes

Design Thinking applied in Business & Strategic Innovation, Design Thinking principles that redefine business - Business challenges: Growth, Profitability, Change, Maintaining Relevance, Extreme competition, Standardisation. Design thinking to meet corporate needs. Design thinking for Startups. Defining and testing Business Models and Business cases. Developing & testing prototypes.

Activity: How to market our own product, about maintenance, reliability and plan for startup.

Textbooks

1. Tim Brown, *Change by design*, 1/e HarperCollins 2008
2. John Hayes, *Design thinking for strategic innovation*, 1/e, John Wiley, 2014

Reference books

1. David Lee, *Design Thinking in the Classroom*, Wynnes press 2018
2. Steven J Sharp, *Design for Future*, 1/e, Norton Press 2012
3. William Lidwell, Kristina Holten, & Jill Butler, *Universal principles of design*, 1/e, Rockport Publishers, 2012
4. Chesbrough, H, *The art of open innovation*, 2003

Online Learning Resources

- + <https://open.ac.uk/courses/112/106/11070016/>
- + <https://open.ac.uk/courses/129/126/12945656/>
- + <https://www.coursera.org/learn/design-thinking>
- + <https://edx.org/course/mitx/6.035.001>

Semester	88	L	T	F	C	COURSE CODE
Regulation	VT2		3		MOE	VT2MTT01
Name of the Course	Professional Communication Skills - I					
Branch	Common to all branches					

Syllabus Details

Course Outcomes

After successful completion of the course, the student will be able to:

CO1: Use vocabulary in regular doses of life with accuracy, make meaningful sentences, and describe people and their traits vividly. (C1)

CO2: Distinguish between places of pilgrimages and holiday spots, describe modern things and process, and frame questions, statements and expressions. (C4)

CO3: Demonstrate their knowledge of idioms which are similar to those of native speakers while speaking and writing and use phrases clearly and precisely to articulate their views that compare and contrast (idioms) with native expressions and avoid common errors. (C4)

CO4: Employ the vocabulary of nations with ease and walk through the letters and emails for effective official correspondence and infer the accurate meaning of the homophones that are often confusing. (C2)

CO5: Summarize their profile, introduce themselves as well as others by incorporating their accomplishments and Youth stories and associated in an interesting and engaging manner that attracts curiosity of the audience. (C4)

UNIT – I

BUILDING VOCABULARY FOR DAILY ACTIVITIES

AIMS: (Fruits, Kitchen Items) – Irrigation- tools – spices- vegetables –flowers – sources of study – Professions

Training Questions – statements – expressions related to the Vocabulary taught

WRITING : Describing people – Physical characteristics- Mental attributes – various professions

Training Questions – statements – expressions related to the Vocabulary taught

ACTIVITY : Related to the topics learnt in Unit – I

No. of hours required – 10

UNIT – II

BUILDING VOCABULARY FOR PLACES, THINGS & PERSONS

PLACES: Describing tourist's place – famous place – Places of Pilgrimage

THINGS - describing a thing - describe an incident or an event

PROCESS - Describe a process - Recipe - experiment - Entrance test application

Forming Questions - Statements - expressions related to the Vocabulary taught

ACTIVITY - Related to the topic learnt in Unit - 1

No. of hours required - 10

UNIT - II

NATIVE EXPRESSIONS - Idioms and Phrases - in day to day activities for different occasions - songs written in English -

PHRASES with as-as expressions - used to compare & contrast

COMMON MISTAKES - in spoken & written

SENTENCES - Most often used expressions - accepted in India - found place in temporary activity - related to the topic learnt in Unit - 1

No. of hours required - 10

UNIT - IV

KEY VOCABULARY - acronyms and abbreviations that are most often used

WORDFORMS - Words often confused - Spelling & Pronunciation

Letter Writing - Formal & Informal - Letters for all occasions

Email Writing - Business mails - protest status mails - informative mails

ACTIVITY - Related to the topic learnt in Unit - 1

No. of hours required - 10

B.Tech CSE (Artificial Intelligence)
&
B.Tech.(Artificial Intelligence & Machine Learning)
V23 IV Semester Course Structure and Syllabus
Course Structure

IV- Semester							
S.No	Course Code	Category	Course	L	T	P	C
1	V23H4007	Engineering Science/ Generic Science	Probability & Statistics	2	0	0	2
2	V23H4009	Management Science-4	Optimization Techniques	2	0	0	2
3	V23H4003	Professional Core	Machine Learning	2	0	0	2
4	V23C0723	Engineering Science	Digital Logic & Computer Organization	2	0	0	2
5	V23C0727	Professional Core	Customer Relationship Systems	2	0	0	2
6	V23H4005	Professional Core	AI & ML Lab	0	0	2	1.5
7	V23C0729	Professional Core	Database Management Systems Lab	0	0	2	1.5
8	V23C03262	Elective Course	Full Stack Developer-1	0	1	2	2
9	V23H4001	Audit Course	Environmental Studies	2	0	0	-
Total				16	01	00	18
<i>Mandatory Community Service Project(V23C03P01) involving of 08 weeks duration during summer vacation</i>							

L-Lecture, T-Practical, P-Project, C-Credits

Syllabus

Year/Sem	Pr. Sem	L	T	P	C	COURSE CODE
Regulation	V22	2	-	-	2	V22MA207
Name of the Course	Probability and Statistics					
Branch	CSE, ECE, CSE (AI), AI/ML					

Syllabus Details

Course Outcome:

Upon completion of the course, the student will be able to

- CO1: Clarify the concept of data science and its importance (K1)
- CO2: Interpret the association of characteristics and through correlation and regression tools (K2)
- CO3: Apply discrete and continuous probability distribution (K3)
- CO4: Construct the components of a classical hypothesis test (K4)
- CO5: Apply the statistical inferential methods to test the hypothesis based on small and large sampling (K5)

UNIT-I: Descriptive statistics and methods for data science:

Data science - Statistics introduction - Population vs Sample - Collection of data - primary and secondary data-type of variable: dependent and independent Categorical and Continuous Variable - Data visualization - Measure of Central tendency - Measure of variability - skewness - Kurtosis

UNIT-II: Correlation and Regression:

Correlation-Correlation coefficient-Rank correlation, Linear Regression: Straight line - Multiple Linear Regression - Regression coefficients and properties - Curvilinear Regression: Parabola - Exponential - Power curves.

UNIT-III: Probability and Distributions:

Probability- Conditional probability and Bayes theorem - Random variables - Discrete and Continuous random variables-Distribution functions-Probability mass function, Probability density function and cumulative distribution function-Statistical Expectation and Variance - Binomial, Poisson, Uniform and Normal distributions, Expectation and Variance-Binomial, Poisson, Uniform and Normal distributions.

UNIT-IV: Sampling Theory:

Introduction - Population and Samples - Sampling distribution of mean and variance (Distribution only)-Point and interval estimates-maximum error of estimate-classical Limit theorem (without proof)-Estimation using χ^2 and F-distributions

Unit 5-V Tests of Hypothesis

Introduction – Hypothesis – Null and Alternative Hypothesis – Type I and Type II errors- Level of significance – One-tail and two-tail tests – Test of significance for large samples and small samples: Single and difference means – Single and two proportions – Student's t-test, F-test, chi-square test.

Text Books:

- Miller and Freund's, *Probability and Statistics for Engineers*, 7th ed, Pearson, 2008.
- R. C. Gupta and V.M. Kapoor, *Fundamentals of Mathematical Statistics*, 11, 12, Sultan Chand & Sons Pvt. Ltd., Meerut, 2012

Reference Books:

- Shree L. Myers, Rayner Va, Ronald E Walpole, *Probability and Statistics Engineers and the Scientists* 8th Edition, Pearson 2007.
- Jay L. Devore, *Probability and Statistics for Engineering and the Sciences*, 8th Edition, Cengage.
- Sheldon M. Ross, *Introduction to probability and statistics Engineers and the Scientists*, 8th Edition, Academic Foundation, 2011.
- Johannes Ledolter and Robert V. Hogg, *Applied statistics for Engineers and Physical Scientists*, 8th Edition, Pearson, 2010.

Year/Sem	IP Sem	L	T	P	C	COURSE CODE
Regulation	V23	3	-	-	6	ME650109
Name of the Course	Optimization Techniques					
Branch	Common to CSE(AI) & AIND					

Syllabus Details

COURSE OBJECTIVES

Upon completion of the course, the students will be able to

- CO1: Formulate the optimization problem for an engineering design problem by using the given constraints and variables (K2)
- CO2: Apply classical optimization techniques to minimize or maximize a multi-variable objective function, without or with constraints (K2)
- CO3: Apply the linear programming methods to solve transportation and assignment problems (K2)
- CO4: Apply gradient and non-gradient methods to nonlinear optimization problems and interior or exterior penalty functions to derive the optimal solution (K3)
- CO5: Apply dynamic programming techniques to inventory control, production planning and engineering design problems for optimal solution (K3)

UNIT I: Introduction and Classical Optimization Techniques

Statement of an Optimization problem, design vector, design constraints, constraint surfaces, objective function, objective function surfaces, classification of optimization problems, Classical Optimization Techniques: single variable Optimization, multi-variable Optimization without constraints, necessary and sufficient conditions for minimum/maximize, multi-variable Optimization with equality constraints, Solution by method of Lagrange multipliers, multi-variable Optimization with inequality constraints, Kuhn - Tucker conditions

UNIT II: Linear Programming

Standard form of a linear programming problem, geometry of linear programming problems, definitions and theorems, solution of a system of linear simultaneous equations, graphical reduction of a general system of equations, introduction to the simplex method, simplex algorithm

UNIT III: Transportation Problems

Finding initial basic feasible solution by north - west corner rule, least cost method and Vogel's approximation method, testing for optimality of balanced transportation problems, Special cases in transportation problem.

UNIT IV: Nonlinear Programming

Unconstrained cases, one - dimensional minimization methods: Classification, Fibonacci method, Golden section method, Steepest descent method. Constrained cases: Characteristics of a constrained problem, Classification, Basic approach of Kuhn-Tucker method, Basic approaches of interior and exterior penalty function methods.

UNIT V: Dynamic Programming

Dynamic programming multi stage decision processes, types, concept of sub optimization and the principle of optimality, computational procedure in dynamic programming, examples illustrating the calculus method of solution, examples illustrating the tabular method of solution.

Text books:

1. "Engineering optimization Theory and practice", S. S. Rao, New Age International (P) limited, 3rd edition, 1998.
2. "Introductory Operations Research", H.S. Kataria & H. Kumar, Springer (India), Pvt. Ltd.

Reference books:

1. Optimization methods in Operations Research and Systems Analysis, by H.S. Kataria and C. Mishra, New Age International (P) Limited, Publishers, Sri Lanka, 1996.
2. Operations Research, Dr. S. D. Sharma, Radhakrishn, Ramnath & Co.

Semester	IV	L	T	P	C	COURSE CODE
Regulation	V23	3	2	0	8	V2341T50
Name of the Course	Machine Learning					
Branch	Common to CSE(AI) and 4150C.					

2) Syllabus Details

Course Outcomes:

After Successful completion of the Course, the student will be able to:

- CO1: Explain various learning techniques and concepts of model. (K1)
- CO2: Demonstrate Nearest Neighbor Based Models (K2)
- CO3: Explain Bayes' Classifier (K2)
- CO4: Demonstrate the concepts of Linear Discriminants and Regression. (K3)
- CO5: Explain various clustering approaches. (K3)

UNIT-I: Introduction to Machine Learning: Evolution of Machine Learning, Paradigms for ML, Learning by rote, Learning by Induction, Reinforcement Learning, Types of Data, Matching, Types of Machine Learning, Data Acquisition, Feature Engineering, Data Representation, Model Selection, Model Learning, Model Evaluation, Model Prediction, Search and Learning, Data Sets

UNIT-II: Nearest Neighbor-Based Models: introduction to Proximity Measures, Distance Measures, Non-Metric Similarity Functions, Proximity Between Binary Patterns, Different Classification Algorithms Based on the Distance Measures, K-Nearest Neighbor Classifier, Radius Distance Nearest Neighbor Algorithm, SVM Regression, Performance of Classifier, Performance of Regression Algorithms

UNIT-III: The Bayes Classifier: Introduction to the Bayes' Classifier, Bayes' Rule and Inference, The Bayes Classifier and its operation, Multi-class classification, Class Conditional Independence and Naïve Bayes Classifier (NBC), Regularization Impact of Regularization in ML, Principal Component analysis Maximum variance formulation, Minimum-error formulation, Applications of PCA, PCA for high-dimensional data

UNIT-IV: Linear Discriminants for Machine Learning: introduction to Linear Discriminants, Linear Discriminants for Classification, Perceptron Classifier, Perceptron Learning Algorithm, Support Vector Machines, Linearly Not Separable Case, Non-linear SVM, Kernel Trick, Logistic Regression, Linear Regression, Multi-Layer Perceptrons (MLPs), Back propagation for Training in MLP

UNIT-V: Clustering : Introduction to Clustering, Partitioning of Data, Matrix Factorization, Clustering of Patterns, Density Clustering, Agglomerative Clustering, Partitional Clustering, K-Means Clustering, Soft Partitioning, Soft Clustering, Fuzzy C-Means Clustering, Rough Clustering, Rough K-Means Clustering Algorithm, Expectation maximization-based Clustering, Spectral Clustering Tutorial.

Textbook:

1. Machine Learning Theory and Practice, H. H. Murthy, V. I. Ananthanarasimhan, Universities Press (India), 2004.

Reference Books:

1. Machine Learning, Tom M. Mitchell, McGraw-Hill Publications, 2017
2. Machine Learning in Action, Peter Harrington, Dream Tech
3. Introduction to Data Mining, Pang-Ning Tan, Michael Steinbach, Fifth Edition, 2018
4. Pattern Recognition and Machine Learning, Christopher M. Bishop, Springer
5. Ulrike Von Luxburg, "A tutorial on spectral clustering", Statistics and computing 17(2007): 393-416

Semester	IV	L	T	P	E	COURSE CODE
Regulation	1/23	3	0	0	3	EEEC0703
Name of the Course	Digital Logic & Computer Organisation					
Branch	Common to ECE(AI) & AIEML					

Syllabus Details

Course Outcomes:

After Successful completion of the Course, the student will be able to:

CO1: Explain different Data Representation and various Combinational Digital Logic Circuits (K2)

CO2: Explain various Sequential Digital Logic Circuits and basic structure of Computers (K2)

CO3: Describe the basic concepts of Computer Arithmetic and Processor Organisation (K2)

CO4: Enumerate different types of Memory and organization of Memory (K2)

CO5: Demonstrate the ways of accessing various interfacing devices with Processor (K2)

UNIT-I:

Data Representations: Binary Numbers, Fixed Point Representation, Floating Point Representation, Signed binary numbers.

Digital Logic Circuits: Basic Logic Functions, Minimization of Logic expressions, K-Map Simplification, Combinational Circuits, Decoders, Multiplexers.

UNIT-II:

Digital Logic Circuits-II: Sequential Circuits, Flip-Flops, Binary counters, Registers, Shift Registers, Ripple counters.

Basic Structure of Computers: Computer Types, Functional units, Basic operational concepts, Bus structures, Software, Performance, Multiprocessors and Multi-computers, Computer Organization, Von-Neuman Architecture.

UNIT-III:

Computer Arithmetic: Addition and Subtraction of signed numbers, Design of Fast Adders, Multiplication of Positive numbers, Signed-squared Multiplication, Fast Multiplication, Integer Division, Floating-Point Numbers and Operations.

Processor Organization: Fundamental Concepts, Execution of a Complete Instruction, Multiple-Bus Organization, Hardwired Control and Micro-programmed Control.

UNIT-IV:

The Memory Organization: Basic Concepts, Semiconductor RAM: Memories, Read-Only Memories, Speed, Size and Cost, Cache Memories, Performance Considerations, Virtual Memories, Memory Management Requirements, Secondary Storage.

UNIT-V:

Input / Output Organization / Addressing I/O Devices, Interrupts, Processor Examples, Direct Memory Access, Buses, Interface Circuits, Standard I/O Interfaces

Textbooks:

1. Computer Organization, Carl Hamacher, Thomas W. Harris, Stuart H. Hartley, 6th edition, McGraw-Hill
2. Digital Design, 6th Edition, M. Morris Mano, Pearson Education

Reference Books:

1. Computer Organization and Architecture, William Stallings, 11th Edition, Pearson.
2. Computer Systems Architecture, M. Hartel, 3rd Edition, Pearson.
3. Computer Organization and Design, David A. Patterson, John L. Hennessy, Elsevier
4. Fundamentals of Logic Design, 6th Edition, Thomson

Online Learning Resources:

1. <https://ojs.uel.br/ojs/bitstream/handle/10007/1001000000/>

Semester	IV	L	T	P	C	COURSE CODE
Regulation	VCO	2	0	0	3	VDCST07
Name of the Course	Database Management Systems					
Branch	Common to CIE, CIT, CSE(A), AISEL and CSD/CG					

Syllabus Details

Course Outcomes

After successful completion of the course, the student will be able to:

CO1: Describe Database systems, various Data models and Database architecture. (K2)

CO2: Develop various real time applications using Relational algebra and Relational calculus (K3)

CO3: Illustrate nested queries and different types of joins. (K4)

CO4: Apply various Normalization techniques to refine schema. (K3)

CO5: Explain Transaction management, concurrency control and Database recovery techniques (K2)

UNIT-1

Introduction: Database system, Characteristics (Database vs File System), Database users, advantages of Database systems, Database applications, Brief introduction of different data Models: concepts of Schema, instance and data independence; Three tier schema architecture for data independence, Database system structure, environment, Centralized and Client Server architectures for the database.

Entity Relationship Model: introduction, Representation of entities, attributes, entity set, relationship, relationship set, constraints, sub classes, super class, inheritance, specialization, generalization using ER Diagrams.

UNIT-2

Relational Model: introduction to relational model, concepts of domain, attribute, tuple, relation, importance of null values, constraints (Domain, Key constraints, integrity constraints) and their importance, Relational Algebra, Relational calculus.

Basic SQL: simple database schema, data types, table definitions (create, alter), different DML operations (insert, delete, update)

UNIT-3

SQL: Basic SQL querying (select and project) using where clause, arithmetic & logical operations, SQL functions (Date and Time, Numeric, String conversion); Creating tables with relationship, implementation of key and integrity constraints, nested queries, sub queries, grouping, aggregation, writing implementation of different types of joins, view (updatable and non-updatable), relational set operations.

UNIT-4

Schema Refinement (normalization): purpose of normalization or schema refinement, concept of functional dependency, normal forms based on functional

dependency based join and dependency preserving decomposition, (1NF, 2NF and 3NF), concept of surrogate key, Boyce-Codd normal form(BCNF), 4NF, Fourth normal form(4NF), Fifth Normal Form (5NF).

UNIT -V

Transaction concept: Transaction state, ACID properties, Concurrency, Deadlock, Serializability, Recoverability, Implementation of database, Tuning for serializability, lock based, time stamp based, optimistic, concurrency protocols, Deadlocks, Failure Classification, Damage, Recovery, and idempotency, Recovery algorithms

Introduction to indexing Techniques: B+ Trees, operations on B+ Trees, Hash Based Indexing

Textbooks:

1. Database Management Systems, 3rd edition, Raghuwama Krishnan, (Johnanes Gabrie, TMH) (For Chapters 2, 3, 4)
2. Database system Concepts, 9th edn, Abraham Silberschatz, Rami Korth, Sudarshan, TMH (For Chapter 1 and Chapter 9)

Reference Books:

1. Introduction to Database Systems, 8th edition, C) Data, Pearson
2. Database Management Systems, 8th edition, Ramon Elmasri, Shamkant D. Navathe, Pearson
3. Database Principles Fundamentals of Design Implementation and Management, Carlos Coronel, Stefan Morris, Peter Robb, Cengage Learning

Online Learning Resources:

1. <https://regular.in/course/186/106/186186176/>
2. <https://techspringboard.com/question/646/646/646186176/>

Semester	IV	L	T	P	C	COURSE CODE
Regulation	V/II	3	0	3	18	VIIIEE21
Name of the Course	AI & ML LAB					
Branch	Common to CSE(AI) and AI&ML					

Syllabus Details

Course Outcomes:

After Successful completion of the Course, the student will be able to:

CO1: Demonstrate the Pandas Library for data manipulation and visualization. (KS)

CO2: Demonstrate Searching techniques. (KS)

CO3: Illustrate the pre-processing approaches.(KS)

CO4: Demonstrate Classification approaches (KS)

Software Required for ML: Python / R / Matlab

List of Experiments

- Pandas Library
 - Demonstrate a Python program to implement Pandas Series with labels.
 - Prepare a Pandas Series from a Dictionary.
 - Prepare a Pandas Data Frame.
 - Illustrate a program which makes use of the following Pandas methods:
 - describe()
 - head()
 - tail()
 - info()
- Pandas Library: Visualisation.
 - Demonstrate a program which use Pandas in built visualization to plot following graphs:
 - Bar plots
 - Histograms
 - Line plots
 - Scatterplots
- Demonstrate a program to Implement Breadth First Search using Python.
- Demonstrate a program to implement Best First Search Algorithm.
- Demonstrate a program to Implement Depth First Search using Python.
- Demonstrate a program to implement the heuristic search.
- Demonstrate a Python program to implement A* and AO* algorithm. (We find the shortest path)
- Apply the following pre-processing techniques for a given dataset.
 - Attribute selection
 - Handling missing values
 - Discretization
 - Min-max or z-transfers

6. Apply K-NN algorithm for classification and regression
7. Demonstrate Decision tree algorithm for a classification problem and perform parameter tuning for better results
8. Apply Random Forest algorithm for classification and regression
9. Demonstrate Naive Bayes Classification algorithm
10. Apply Support Vector algorithm for classification
11. Illustrate the K-means algorithm and apply it to the data you collected. Evaluate performance by measuring the sum of the Euclidean distances of each sample from its data center. Test the performance of the algorithm as a function of the parameter k .

Reference Books

1. Machine Learning Theory and Practice, H. H. Maathly, V. L. Ananthanarayanan, Universities Press (India), 2014
2. Machine Learning Tom M. Mitchell, McGraw-Hill Education, 2017
3. Machine Learning in Action, Peter Harrington, No Starch Press
4. Introduction to Data Mining, Pang-Ning Tan, Michael Steinbach, Vijay Kumar, 7th Edition, 2019.
5. Pattern Recognition and Machine Learning, Christopher M. Bishop, Springer

Semester	IV	L	T	P	C	COURSE CODE
Regulation	V23	3	0	3	1.5	V23CSL57
Name of the Course	Database Management Systems Lab					
Branch	Common to CSE, ECE, EEE(A), AIMSL and CEE(DC)					

Syllabus Details

COURSE OBJECTIVES

After Successful completion of the Course, the student will be able to:

- ED1: Construct SQL queries using DDL, DML and DCL commands. (N3)
- ED2: Experiment with nested queries, built-in functions and various constraints. (N2)
- ED3: Construct PL/SQL programs as control structures, functions and Triggers. (N3)
- ED4: Develop a Java program to connect with a database using JDBC. (N2)

Experiments covering the Topics:

- DDL, DML, DCL commands
- Queries, nested queries, built-in functions
- PL/SQL programming- control structures
- Procedures, Functions, Cursors, Triggers
- Database connectivity- JDBC, JSPC

Sample Programs:

1. Demonstrate creation, altering and dropping of tables and inserting rows into a table (use constraints while creating tables) examples using SELECT command.
2. Construct queries (along with sub queries) using ANY, ALL, IN, EXISTS, NOT EXISTS, UNION, INTERSECT, Constraints. Example- Select the roll number and name of the student who scored fourth rank in the class.
3. Construct queries using Aggregate functions (COUNT, SUM, AVE, MAX and MIN), GROUP BY, HAVING and Creation and dropping of Views.
4. Construct queries using Conversion functions (to_char, to_number and to_date), string functions (Concatenation, lpad, rpad, trim, rtrim, lower, upper, initcap, length, substr and substrb), date functions (sysdate, next_day, add_months, last_day, months_between, least, greatest, trunc, round, to_char, to_date).
5.
 - a. Develop a simple PL/SQL program which includes declaration section, executable section and exception -Handling section [Ex: Student marks can be selected from the table and printed for those who scored first class and an exception can be raised if no records were found]
 - b. Insert data into student table and use COMMIT, ROLLBACK and SAVEPOINT in PL/SQL block.
 - c. Develop a program that includes the features BULK COLLECT and FORALL.

expression. The program can be extended using the `SQLERRM` and `SQLERRN` functions.

1. Develop programs using `WHILE` LOOPS, numeric `FOR` LOOPS, nested loops using `EXCEPTION` Handling, `BUILT-IN` Exceptions, `USER` defined Exceptions, `Raise-APPLICATION ERROR`.
2. Develop programs using procedures, passing parameters in and out of `PROCEDURES`.
3. Develop programs on stored functions, inline functions in `SQL`, `Anonymous` and `Complex` functions.
4. Develop programs using function parameters in a `CURSOR`, `FOR UPDATE CURSOR`, `WHERE CURRENT OF` clause and `CURSOR` variables.
5. Develop Programs using `BEFORE` and `AFTER` Triggers, `Row` and `Statement` Triggers and `INSTEAD OF` Triggers.
6. Construct a table and perform the search operation on table using indexing and non-indexing techniques.
7. Develop a Java program that connects to a database using `JDBC`.
8. Develop a Java program to connect to a database using `JDBC` and insert values into it.
9. Develop a Java program to connect to a database using `JDBC` and delete values from it.

Textbooks:

1. Database Management Systems, 3rd edition, Raghuwamini Kortham, (John Wiley & Sons, 1998) (For Chapters 2, 3, 4)
2. Database System Concepts, 6th edition, Abraham Silberschatz, Rami Abraham, 1998 (For Chapter 1 and Chapter 3)

Reference Books:

1. Oracle: The Complete Reference by Oracle Press
2. Nilash Datta, "Database Systems Using Oracle", PHI, 2007
3. Rick F Vander Laan, "Introduction to SQL", Fourth Edition, Pearson Education, 2007

Semester	IV	L	T	F	C	COURSE CODE
Regulation	VCS	3	1	2	2	VCS030302
Name of the Course	Full Stack Development-I					
Branch	Common to CSE, CST, CSE(40) and AIML					

Syllabus Details

Course Outcomes

After successful completion of the Course, the student will be able to:

- CO1: Identify HTML elements and their attributes for designing static web pages. (K1)
- CO2: Apply appropriate CSS styles to HTML elements. (K2)
- CO3: Demonstrate JavaScript Pre-defined and User-defined Objects. (K3)
- CO4: Develop dynamic web pages and validate forms using JavaScript. (K4)

Experiments covering the Topics:

- Lists, Links and Images
- HTML Tables, Forms and Frames
- HTML 5 and Cascading Style Sheets: Types of CSS
- Selector forms
- CSS with Color, Background, Font, Text and CSS Box Model
- Applying JavaScript: internal and external, I/O, Type Conversion
- JavaScript Conditional Statements and Loops, Pre-defined and User-defined Objects
- JavaScript Functions and Events
- Node.js

Sample Programs:

1. Lists, Links and Images
 - Develop a HTML program, to explain the working of lists.
 - name. It should have an ordered list, unordered list, nested list and ordered list as an unordered list and definition list.
 - Develop a HTML program, to explain the working of hyperlinks using <a> tag and href, target attributes.
 - Develop a HTML document that has your image and your friend's image with a specific height and width. Also when clicked on the images it should navigate to their respective profiles.
 - Develop a HTML program, in such a way that, rather than placing large images on a page, the preferred technique is to use thumbnails by setting the height and width parameters to something like 100*100 pixels. Each thumbnail image is also a link to a full sized version of the image. Create an image gallery using this technique.
2. HTML Tables, Forms and Frames
 - Develop a HTML program, to explain the working of tables (see tags: <table>, <tr>, <td>, <th>, <caption>, border, colspan, rowspan)
 - Develop a HTML program, to explain the working of tables by preparing a

invisible. (Note: Use <caption> tag to set the caption to the table & also use cell spacing, cell padding, border, rowspan, colspan etc.)

- ii) Develop a HTML program, to explain the working of forms by designing Registration form. (Note: Include text field, password field, number field, date of birth field, checkboxes, radio buttons, list boxes using <select> & <option> tags, <input type="button"> and two buttons to submit and reset. Use tables to provide a better view).
- iii) Develop a HTML program, to explain the working of frames, such that page is to be divided into 3 parts or either direction. (Note: first frame is image, second frame is paragraph, third frame is hyperlink and also make use of using "no frame" attribute such that frames to be closed).

3. HTML 5 and Cascading Style Sheets, Types of CSS

- i) Develop a HTML program, that makes use of <video>, <audio>, <figure>, <figcaption>, <chart>, <table>, <caption>, <code>, <math>, <div>, , <sup>, <sub> tags.
- ii) Develop a HTML program, to embed audio and video into HTML web page.
- iii) Develop a program to apply different types (or levels of style or style specification format)- inline, internal, external styles to HTML elements (identify selector, property and value).

A. Selector forms

- i) Develop a program to apply different types of selector forms
 - a. Single selector (element, id, class, group, universal)
 - ii. Combinator selector (descendant, child, adjacent sibling, general sibling)
 - iii. Pseudo-class selector
 - iv. Pseudo-element selector
 - v. attribute selector

3. CSS with Color, Background, Font, Text and CSS Box Model

- i) Develop a program to demonstrate the various ways you can reference a color in CSS.
- ii) Develop a CSS rule that places a background image halfway down the page, aligning it horizontally. The image should remain in place when the user scrolls up or down.
- iii) Develop a program using the following names related to CSS font and text:
 - a. font-size ii. font-weight iii. font-style
 - iv. text-decoration v. text-decoration vi. text-align
- iv) Develop a program, to explain the importance of CSS Box model using
 - i. Content ii. Border iii. Margin iv. Padding

4. Applying JavaScript - Internal and external, I/O, Type Conversion

- i) Develop a program to embed internal and external JavaScript in a web page.
- ii) Develop a program to explain the different ways for displaying output.
- iii) Develop a program to explain the different ways for taking input.
- iv) Create a webpage which uses prompt dialog box to ask a voter for his age and age, display the information in table format along with whether the voter can vote or not.

7. JavaScript Pre-defined and User-defined Objects

- i) Develop a program using document object properties and methods.
- ii) Develop a program using window object properties and methods.
- iii) Develop a program using array object properties and methods.
- iv) Develop a program using math object properties and methods.
- v) Develop a program using string object properties and methods.
- vi) Develop a program using regex object properties and methods.
- vii) Develop a program using date object properties and methods.
- viii) Develop a program to explain user-defined object by using properties, methods, accessors, constructors and display.

8. JavaScript Conditional Statements and Loops

- i) Develop a program which asks the user to enter three integers, obtains the numbers from the user and outputs HTML text that displays the larger number followed by the words "LARGER NUMBER" in an information message dialog. If the numbers are equal, output HTML text as "DYNAMIC NUMBER".
- ii) Develop a program to display week days using switch case.
- iii) Develop a program to print 1 to 10 numbers using for, while and do-while loops.
- iv) Develop a program to print data in object using form, foreach and for-of loops.
- v) Develop a program to determine whether a given number is an "ARMSTRONG NUMBER" or not [eg. 153 is an Armstrong number, since sum of the cube of the digits is equal to the number i.e. $1^3 + 5^3 + 3^3 = 153$].
- vi) Develop a program to display the denomination of the amount deposited in the bank in terms of 100's, 50's, 20's, 10's, 5's, 2's & 1's (eg. if deposited amount is Rs.165, the output should be 1-100's, 1-00's, 1-10's, 1-0's & 1-1's).

9. JavaScript Functions and Events

- i) Design a appropriate function should be called to display:
 - a. Factorial of that number
 - b. Fibonacci series up to that number
 - c. Prime numbers up to that number
 - d. Is it palindrome or not
- ii) Prepare a HTML file having a text box and four buttons named Factorial, Fibonacci, Prime, and Palindrome. When a button is pressed an appropriate function should be called to display:
 - a. Factorial of that number
 - b. Fibonacci series up to that number
 - c. Prime numbers up to that number
 - d. Is it palindrome or not
- iii) Develop a program to validate the following fields in a registration page:
 - a. Name (start with alphabet and followed by alphanumeric and the length should not be less than 8 characters)
 - b. Mobile (only numbers and length 10 digits)
 - c. E-mail (should contain format like xxxxxxx@xxxxxx.xx)

Reference Books:

1. Programming the World Wide Web, 4th Edition, Robert W. Stevens, Pearson, 2018.
2. Web Programming with HTML5, CSS and JavaScript, John Dean, Jonas & Damien Larmann, 2019 (Chapters 1-11).
3. Full Stack Node: Full Stack Node App Development with MongoDB, Express, React, and Node, Hassan Subramanian, 2nd edition, A Press, O'Reilly.

Online Learning Resources:

1. <https://www.w3schools.com/html/>
2. <https://www.w3schools.com/css/>
3. <https://www.w3schools.com/js/>
4. <https://www.w3schools.com/nodejs/>

Semester	IV	L	T	P	C	COURSE CODE
Regulation	VIZ	2	0	0	-	VIZCE4001
Name of the Course	Environmental Science					
Branch	Common to CSE, CTE, CSE(A) and A/BML					

3) Syllabus Details

Course Outcomes:

After Successful completion of the Course, the student will be able to:

- CO1: Delineate multi-disciplinary nature of environmental studies and various renewable and non-renewable resources. (K1)
- CO2: Define flow and bio-geo-chemical cycles and ecological Pyramids. (K4)
- CO3: Ascertain various sources of pollution and solid waste management and related preventive measures. (K2)
- CO4: Delineate the rain water harvesting, watershed management, ozone layer depletion and waste land reclamation. (K4)
- CO5: Illustrate the causes of population explosion, value education and welfare programs. (K3)

UNIT-1: Multidisciplinary Nature of Environmental Studies: - Definition, Scope and Importance - Need for Public Awareness, Natural Resources: Renewable and non-renewable resources - Natural resources and associated problems - Forest resources - Use and over-exploitation, deforestation, case studies - Timber extraction - Mining, dams and other effects on forest and tribal people - Water resources - Use and over utilization of surface and ground water - Floods, drought, conflicts over water, dams - benefits and problems - Mineral resources: Use and exploitation, environmental effects of extracting and using mineral resources, case studies - Food resources: World food problems, changes caused by agriculture and overgrazing, effects of modern agriculture, fertilizer-pesticide problems, water logging, salinity, case studies - Energy resources

UNIT-2: Ecosystems: Concept of an ecosystem - Structure and function of an ecosystem - Producers, consumers and decomposers - Energy flow in the ecosystem - Ecological succession - Food chains, food webs and ecological pyramids - Introduction, types, characteristic features, structure and function of the following ecosystem:

- a. Forest ecosystem
- b. Grassland ecosystem
- c. Desert ecosystem
- d. Aquatic ecosystems (ponds, streams, lakes, rivers, oceans, estuaries)

Biodiversity and its Conservation: Introduction and Definition: genetic, species and ecosystem diversity - Bio-geographical classification of India - Value of biodiversity: consumptive use, productive use, social, ethical, aesthetic and

option values – Biodiversity at global, national and local levels – India as a mega-diversity nation – Hot-spots of biodiversity – Threats to biodiversity: habitat loss, poaching of wildlife, man- wildlife conflicts – Endangered and endemic species of India – Conservation of biodiversity: In situ and Ex situ conservation of biodiversity

UNIT-III: Environmental Pollution: Definition, Causes, effects and control measures of:

- a. Air Pollution
- b. Water pollution
- c. Soil pollution
- d. Marine pollution
- e. Noise pollution
- f. Thermal pollution
- g. Nuclear hazards

Solid Waste Management: Causes, effects and control measures of urban and industrial wastes – Role of an individual in prevention of pollution – Pollution case studies – Disaster management: floods, earthquakes, cyclones and landslides

UNIT-IV: Social Issues and the Environment: From unsustainable to sustainable development – Urban problems related to energy – Water conservation, rain water harvesting, watershed management – Resettlement and rehabilitation of people: its problems and concerns. Case studies – Environmental effects: issues and possible solutions – Climate change, global warming, acid rain, ozone layer depletion, nuclear accidents and holocaust. Case studies- waste land reclamation-Composures and waste products- Environment Protection Act – Air (Prevention and Control of Pollution) Act – Water (Prevention and control of Pollution) Act – Wildlife Protection Act – Forest Conservation Act – Issues involved in enforcement of environmental legislation – Public awareness.

UNIT-V: Human Population And The Environment: Population growth, variation among nations. Population explosion – Family Welfare Programmes – Environment and human health – Human Rights – Value Education – HIV/AIDS – Women and child Welfare – Role of information Technology in environment and human health – Case studies: Field Work: Visit to a local area to document environmental assets: River/forest/grassland/hill/ wetland – Visit to a local polluted site: Urban/Rural/Industrial/agricultural study of common plants, insects and birds – river, hill slopes, etc.

Textbooks:

1. Drack Ghosh, Textbook of Environmental Studies for Undergraduate Courses, Universities Press (India) Private Limited, 2010.
2. Palaniswamy, Environmental Studies, 2nd Ed., Pearson Education, 2014.
3. Assem Unnisa S, Environmental Studies, Academic Publishing Company, 2011.
4. Naghvan Marbaf, E. Text book of Environmental Studies for Undergraduate courses as per UGC model syllabus, TarTech Publications (India) Pvt. Ltd, 2010.

Reference Books

1. Desikha Devi and E.S. Saba Reddy, Textbook of Environmental Science, 2nd Ed. Cengage Publications, 2012
2. Ang Reddy M. Textbook of Environmental Science and Technology, 3rd Edition, 2014.
3. Sharma J.P. Comprehensive Environmental Studies, Laxmi Publications, 1999
4. Gyere Henry J. and Gary W. Munro, Environmental Science and Engineering, Prentice Hall of India Private Limited

Semester	IV	L	T	P	C	COURSE CODE
Regulation	NZS	-	2	-	-	VSSE203
Name of the Course	Professional Communication Skills-II					
Branch	Common to CSE, CDT, CIB&IT and AIB&I					

Course Details

Course Description:

After successful completion of the Course, the student will be able to:

- CO1: Define the meaning and significance of Soft Skills in contrast to Hard Skills and English, and learn how to enhance them (K1).
- CO2: Identify INOT and demonstrate soft skills competency in the workplace through intrapersonal skills (K1)
- CO3: Demonstrate interpersonal skills by displaying leadership and team work in the working environment for professional challenges (K1)
- CO4: Employ oral communication skills effectively and have an appreciable demeanor in the workplace(K2)
- CO5: Proficiently display skills effectively, meeting the requirements of the corporate world (K2)

UNIT - 1: Introduction

Definition & Meaning-Importance, Soft skills vs. hard skills, Soft skills vs. English-Improving Techniques

UNIT - II: Intra-Personal

Definition & Meaning- Importance - INOT analysis - Goal Setting- Emotional Intelligence-Attitudinal skills- Problem Solving- Time Management- Stress management

UNIT-III: Inter-Personal

Definition & Meaning-Importance-Communication skills-Teamwork, Managerial skills-Negotiation skills-Leadership skills-Corporate etiquette

UNIT - IV: Oral Communication Skills (L&P) & Kinesthetic

Listening: Importance-Improving Tips for Listening & Practice

Kinesthetic: Eye contact, Facial expressions, Gestures, Postures, Appearance, and Dress code.

Speaking: Importance and Supra segmental Features, Delivery and Behavioral Interview.

UNIT-V Literary Skills(R&W)

Reading: Strategies, Reading Comprehensions (Literal and Inferential)
Writing: Technical Report, Passages with Cover Letter.

Textbooks:

- 1) Shakti K. M. Kulkarni, *Corporate Developing Soft Skills*,/s. Pearson Publications, New Delhi, 2014.
- 2) Allen Walker, *Life Skills for Success*, L/s Sage Publications India Private Limited, 2016.

Reference Books

1. Samirash M. Technical English, Wiley publishers India New Delhi, 2014.
2. Gangadhar Joshi, *From Campus to Corporate*, SAGE TEXT.
3. Alex K. Soft Skills, 3rd ed. S. Chand Publication, New Delhi, 2014.
4. Meenakshi Rameshwar Dangele Sharma, *Technical Communication: Principles and Practice*, Oxford University Press, 2009.
5. Shashi Varma, *Body Language for Your Success*, Harra/s, S. Chand Publication, New Delhi, 2014.
6. Stephen Covey, *Seven Habits of Highly Effective People*, J H D Book, 2012.

Online Learning Resources

- https://oercommons.org/catalog/view.php?record_id=160&preview
- <http://www.youtube.com/watch?v=4w4p0m2T11E>
- https://www.youtube.com/watch?v=74Ly_28G0870Q&list=PLC0b0d8T_g5000_g4v-w500085D-076287k
- <https://www.coursera.org/learn/people-soft-skills-courseware/lecture/english>
- <https://www.coursera.org/learn/soft-skills>